Card column reference

All **711** columns across the five cards, grouped by card and then by block, with blocks ordered by **measured importance**. Each entry leads with what the column *is*; the caveats fold underneath.

Generated 2026-08-20 by `analyses/ops/gen_column_reference.py` from `card_glossary.tsv` and `card_registry.tsv`. **Do not hand-edit.**

Vintage mix â€” read before quoting a fitted column

The delivered edge card is **16 days newer** than the base it layers over. Its **84 base-owned columns** (`coupling_`, `beta_`, `dose_`, `share_`, `rank_`) carry the older vintage; the other 102 are current.

The one rule behind this layout

The rung is a property of (CARD, COLUMN) â€” never of the column name. `beta` is edge-rung on the edge card and family-rung on the gene_family card: same estimator, different unit. That is why this is organised card-first, and why one name can appear twice with different text.

How blocks are ordered

By three measured signals, not by size: how often a block's columns are named in the **discovery registry** (what the findings rest on, weighted highest), how many **modules** read them, and whether the block is **base-owned** (fitted) rather than an annotation join. Names under 6 characters count only backtick-quoted registry hits â€” otherwise `n` and `z` match ordinary prose.

One correction the measurement needed: mention-frequency is not importance â€” a *settled* block gets discussed less precisely because nobody argues about it. `adm_` scored 27th of 44 on the raw signals while MH-216 calls `adm_has_site` the single most load-bearing conditioning variable, on 4 registry mentions. Five blocks therefore carry a **curated floor**, each citing the finding that justifies it (see `CURATED_FLOOR`). That is the only hand-set input here.

Cards

- `edge â€” 186 columns, 47 blocks`
- `gene â€” 135 columns, 29 blocks`
- `gene_family â€” 61 columns, 12 blocks`
- `arm â€” 296 columns, 44 blocks`
- `seed_family â€” 33 columns, 2 blocks`

realization/edge_card.tsv

edge

One row per one (miRNA arm, gene) pair. The widest card and the one most claims are made from. 84 of its columns come from the BASE card and 102 from the annotation layer â€” check the vintage banner before quoting a fitted one.

186 columns **5,649** rows **47** blocks median coverage **72%**

Contents â€" 47 blocks, most important first

(BARE) Â· 6

core â€" the columns findings rest on

6 columns Â· **spans edge, key** â€" this prefix groups columns living on DIFFERENT units Â· coverage **96%â€"100%** (median 100%)
2× identifier Â· 2× real â‰¥ 0 Â· 1× boolean Â· 1× real, signed

arm
KEY IDENTIFIER 100%

KEY â€" mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED

example: `CDH1 / miR-25-3p/32-5p/92-3p/363-3p/367-3p â†' hsa-miR-25-3p`

beta
EDGE REAL â‰¥ 0 100%

EDGE-rung coupling coefficient â€" `readouts.run(level='arm')`, the same-seed collapse REMOVED. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.2203`

gene
KEY | IDENTIFIER | 100%

KEY â€" HGNC gene symbol. AUTHORED
example: `hsa-miR-21-5p / miR-21-5p/590-5p â†' ACAT1`

---

identified
EDGE | BOOLEAN | 100%

`|z| > 2` on the UNCALIBRATED posterior SD, where `z = beta / beta_sd` (MH-83 records precision 0.86 / recall 0.89 against a held-out standard). **True on 24.8% of edges.** AUTHORED
values: `False (4,269) Â· True (1,375)`
example: `CDH1 / hsa-miR-25-3p â†' True`

---

identity
EDGE | REAL, SIGNED | 96%

Shapley/LMG share of the gene's R^2 credited to this edge, with bagged-NNLS weights. AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' 0.3367`

---

z
EDGE | REAL â‰¥ 0 | 100%

`beta / beta_sd` for the EDGE-level fit (`readouts.run(level='arm')`, the whole gene's arms as separate predictors), on the UNCALIBRATED posterior SD. AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' 4.542`

COUPLING_ Â· 13

core â€" the columns findings rest on

> Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
>
> **13** columns Â· **spans edge, family** â€" this prefix groups columns living on DIFFERENT units Â· coverage **3%â€"100%** (median 62%) Â· **9** share the block description
>
> 5× signed âˆ'1â€¦1 Â· 5× p-value Â· 3× real, signed

coupling_fam
FAMILY | SIGNED âˆ'1â€¦1 | 100%

Family-level coupling for the edge's (gene, seed_family) cell. AUTHORED
example: `CDH1 / hsa-miR-25-3p â†' -0.2577`

---

coupling_hly
EDGE | SIGNED âˆ'1â€¦1 | 56%

in the GTEx-healthy state DERIVED
example: `CDH1 / hsa-miR-25-3p â†' 0.007`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

---

coupling_hly_surrogate
EDGE | SIGNED âˆ'1â€¦1 | 3%

Healthy coupling via a same-seed SURROGATE arm, used where the arm itself has no GTEx signal. AUTHORED
example: `CDH1 / hsa-miR-30a-5p â†' -0.037`

---

coupling_nat
EDGE | SIGNED âˆ'1â€¦1 | 62%

in matched normal (NAT) DERIVED
example: `CDH1 / hsa-miR-25-3p â†' -0.116`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

coupling_p_hly
EDGE | P-VALUE | 56%

p-value Â· in the GTEx-healthy state DERIVED
example: `CDH1 / hsa-miR-25-3p â†' 0.6225`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

---

coupling_p_hly_resolved
EDGE | P-VALUE | 59%

p-value Â· in the GTEx-healthy state DERIVED
example: `CDH1 / hsa-miR-25-3p â†' 0.6225`

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€" 3.6% of edges*

---

coupling_p_hly_surrogate
EDGE | P-VALUE | 3%

p-value Â· in the GTEx-healthy state DERIVED
example: `CDH1 / hsa-miR-30a-5p â†' 0.4457`

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€" 3.6% of edges*

coupling_p_nat
EDGE | P-VALUE | 62%

p-value Â· in matched normal (NAT) DERIVED

example: CDH1 / hsa-miR-25-3p â†’ 0.1658

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

coupling_p_tum
EDGE | P-VALUE | 69%

p against the CALIBRATED site-free null (not a theoretical t-null). Median 0.375; <0.05 on 16.2%. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.0012

---

coupling_tum
EDGE | SIGNED Â^'1Â€¦1 | 72%

in tumour DERIVED

example: CDH1 / hsa-miR-25-3p â†’ -0.271

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

coupling_z_hly
EDGE | REAL, SIGNED | 59%

z-score Â· in the GTEx-healthy state DERIVED

example: CDH1 / hsa-miR-25-3p â†’ 0.31

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)*

---

coupling_z_nat
EDGE | REAL, SIGNED | 62%

z-score Â· in matched normal (NAT) DERIVED

example: CDH1 / hsa-miR-25-3p â†’ -0.97

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

coupling_z_tum
EDGE | REAL, SIGNED | 69%

The tumour coupling expressed in NULL SDs â€” the CALIBRATED effect size, and the scale to quote. Median **â^'0.32**; |z|>2 on 15.5%, |z|>3 on 6.4%. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ -3.04

SEED_ Â· 1

core â€” the columns findings rest on

1 columns Â· rung **family** Â· coverage **100%â€“100%** (median 100%)
1× identifier

seed_family
FAMILY | IDENTIFIER | 100%

KEY â€” the seed family itself, one row per family. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ miR-25-3p/32-5p/92-3p/363-3p/367-3p

---

ADM_ Â· 4

core â€” the columns findings rest on

4 columns Â· rung **edge** Â· per-edge FLAGS; the arm card carries their rollup Â· coverage **87%â€“87%** (median 87%)
4× boolean

adm_admissible
EDGE | BOOLEAN | 87%

The admissibility verdict: site + evidence + expression. AUTHORED

values: True (2,981) Â· False (1,956)

example: CDH1 / hsa-miR-25-3p â†’ True

---

adm_expr_gtex
EDGE | BOOLEAN | 87%

ADMISSIBILITY â€” could this edge repress in breast AT ALL? `has_site` (seed match in the 3'UTR) AND `expressed` (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED

values: False (3,927) Â· True (1,010)

example: CDH1 / hsa-miR-25-3p â†’ True

---

adm_expr_tcga
EDGE | BOOLEAN | 87%

Is the arm expressed above floor in TCGA? AUTHORED

values: True (3,154) Â· False (1,783)

example: CDH1 / hsa-miR-25-3p â†’ True

adm_has_site

EDGE | BOOLEAN | 87%

Does the arm have a seed match in this target's 3'UTR. AUTHORED

values: `True (4,436) Â· False (501)`

example: `CDH1 / hsa-miR-25-3p â†' True`

ARM_ Â· 15

core â€" the columns findings rest on

> **15** columns Â· **spans arm, arm-in-family, family** â€" this prefix groups columns living on DIFFERENT units Â· coverage **20%â€"100%** (median 69%)
> 4× real â‰¥ 0 Â· 4× real, signed Â· 3× fraction 0â€"1 Â· 2× categorical Â· 1× percent Â· 1× boolean

arm_credit_share

ARM-IN-FAMILY | FRACTION 0â€"1 | 20%

This arm's share of its (gene, seed_family) cell's credit. âœ... Verified to sum to 1.0000 within every one of the 467 multi-arm cells, and it genuinely varies PER ARM inside the cell (467/467). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.7094`

> *Defined on: multi-arm cells only (n_arm_in_cell > 1) â€" 20.3% of edges*

---

arm_dbeta

FAMILY | REAL â‰¥ 0 | 20%

maxâˆ'min Î² across the cell's member arms. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.03`

---

arm_detection

ARM | FRACTION 0â€"1 | 95%

Fraction of patients in which this ARM is measured at all (`readouts._detection`). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 1`

---

arm_id_status

ARM-IN-FAMILY | CATEGORICAL | 100%

How the arm's identity within its cell was resolved: `resolved_by_dose` 2,344 Â· `singleton` 2,260 Â· `unvalidated_balanced` 1,045. AUTHORED

values: `resolved_by_dose (2,344) Â· singleton (2,260) Â· unvalidated_balanced (1,045)`

example: `CDH1 / hsa-miR-25-3p â†' unvalidated_balanced`

---

arm_iqr

ARM | REAL â‰¥ 0 | 69%

Interquartile range of the arm's abundance â€" its DYNAMIC RANGE. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.9`

---

arm_lfc_HLY_NAT_raw

ARM | REAL, SIGNED | 63%

log-FC GTEx-healthy â†' TCGA-NAT, raw. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 9.49`

---

arm_lfc_HLY_TUM_QN

ARM | REAL, SIGNED | 64%

log-FC GTEx-healthy â†' TCGA-tumour, on quantile-normalised values. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 4.79`

---

arm_lfc_HLY_TUM_raw

ARM | REAL, SIGNED | 63%

As `arm_lfc_HLY_TUM_QN` but against the RAW GTEx median. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 9.46`

---

arm_lfc_NAT_TUM

ARM | REAL, SIGNED | 68%

log-FC of the arm's median abundance, matched NAT â†' tumour. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.03`

---

arm_med_rpm

ARM | REAL â‰¥ 0 | 69%

Median linear RPM of this ARM across the cohort. Arm-rung, so it REPEATS across the arm's genes. Fill 0.686 â€" the gap is arms with no expression record, not zeros. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 13.1`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_pct_above_floor

ARM | PERCENT | 69%

Percent of samples in which the arm sits above the detection floor (median 99). Feeds `arm_spiker` together with `arm_iqr` . AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 100`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

arm_role_in_family
ARM-IN-FAMILY · CATEGORICAL · 72%

â›” **The ARM's DOSE role inside its seed family â€” NOT the family's role in the gene**, which is what the name suggests. AUTHORED

values: `sole (2,260) Â· co (1,045) Â· minor (395) Â· dominant (379)`

example: `CDH1 / hsa-miR-25-3p â†' co`

---

arm_sep_z
FAMILY · REAL Â‰¥ 0 · 20%

`arm_dbeta` in units of the cell's own pooled posterior SD â€” can these same-seed arms be told apart at all. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 3.196`

---

arm_share_of_family_dose
ARM-IN-FAMILY · FRACTION 0Â€“1 · 69%

The arm's share of its seed family's TOTAL dose â€” denominator is the **full family**, not the design (median 0.69). The quantity `family_role` is cut from. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.56`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_spiker
ARM · BOOLEAN · 69%

`arm_pct_floor < 40 AND arm_iqr > 1.5` â€” the arm sits at/below the detection floor in most samples yet swings widely when present. AUTHORED

values: `False (3,706) Â· True (171)`

example: `CDH1 / hsa-miR-25-3p â†' False`

BETA_ Â· 8

core â€” the columns findings rest on

> **8** columns Â· rung **edge** Â· coverage **20%â€“100%** (median 100%)
> 4× real â‰¥ 0 Â· 4× fraction 0â€“1

beta_arm
EDGE · REAL Â‰¥ 0 · 20%

The arm's OWN Î² from the within-cell fit (`arm_rung.py`), defined only in multi-arm cells (fill 0.203). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.04066`

---

beta_deconv
EDGE · REAL Â‰¥ 0 · 100%

Î² refit on the DECONV design (core C + the 8 non-malignant Wu-major lineages, Cancer-Epithelial held out). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.1737`

---

beta_frac
EDGE · FRACTION 0Â€“1 · 100%

MAGNITUDE share (beta_f / sum beta). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.1575`

---

beta_frac_abs
EDGE · FRACTION 0Â€“1 · 100%

|E[Î²_f]| / Î£|E[Î²]| â€” the share computed from the POINT ESTIMATES. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.1565`

---

beta_frac_deconv
EDGE · FRACTION 0Â€“1 · 100%

`beta_frac` recomputed on the deconv design â€” the composition-adjusted share of the gene's budget. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.1411`

---

beta_frac_sd
EDGE · FRACTION 0Â€“1 · 100%

Posterior SD of the per-draw fraction Î²_f/Î£Î². AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.03573`

---

beta_sd
EDGE · REAL Â‰¥ 0 · 100%

Posterior summaries of the coupling coefficient from the dense Gibbs fit (`learned/readouts.py`). `_sd` is the posterior SD, `_frac` the share of the gene's total beta. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.04851`

---

beta_sd_deconv
EDGE · REAL Â‰¥ 0 · 100%

Posterior SD on the deconv design. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.04548`

N_ Â· 5

core â€” the columns findings rest on

5 columns Â· **spans edge, family, gene** â€” this prefix groups columns living on DIFFERENT units Â· coverage **20%â€“100%** (median 100%)
4× count Â· 1× constant

n_HLY_meas
EDGE COUNT 72%

How many of this gene's arms have a MEASURED healthy (GTEx) level â€” as opposed to one manufactured by the multi-mapping collapse. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 8

n_arm_in_cell
FAMILY COUNT 20%

How many arms sit in this edge's (gene, seed_family) CELL â€” i.e. how many of that family's members appear in THIS gene's design. AUTHORED

values: 2.0 (670) Â· 3.0 (258) Â· 4.0 (100) Â· 5.0 (55) Â· 6.0 (36) Â· 7.0 (28)

example: CDH1 / hsa-miR-25-3p â†' 2

n_fam
GENE COUNT 100%

How many families compete at this edge's GENE. Gene-rung â€” it repeats across the gene's edges by construction. beta is mathematically inert where it is 1. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 25

n_family_in_design
FAMILY COUNT 100%

Members of this arm's seed family **THAT ARE IN THIS GENE'S DESIGN** â€” 1 on **4,497 of 5,644 edges (79.7%)**. AUTHORED

values: 1.0 (4,497) Â· 2.0 (670) Â· 3.0 (258) Â· 4.0 (100) Â· 5.0 (55) Â· 6.0 (36) Â· 7.0 (28)

example: CDH1 / hsa-miR-25-3p â†' 2

n_samples
GENE CONSTANT 100%

Samples the edge was fit on â€” CONSTANT (1,040) across the card. AUTHORED

values: 1040.0 (5,644)

example: CDH1 / hsa-miR-25-3p â†' 1040

SHARE_ Â· 4

core â€” the columns findings rest on

The arm's share of the gene's total regulatory dose in one state (HLY / NAT / TUM). Within-gene, sums to 1 across the gene's arms.

4 columns Â· rung **edge** Â· coverage **46%â€“72%** (median 72%) Â· **2** share the block description
4× fraction 0â€“1

share_HLY
EDGE FRACTION 0â€“1 72%

The arm's share of the gene's healthy (GTEx) pressure budget, median **0.129**. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.15

share_HLY_meas
EDGE FRACTION 0â€“1 46%

`share_HLY` over ONLY the arms GTEx genuinely measures â€” invisible arms ABSTAIN (NaN) instead of being zeroed (MH-210). Median **0.244** against the imputed 0.129. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.297

share_NAT
EDGE FRACTION 0â€“1 72%

in matched normal (NAT) DERIVED

example: CDH1 / hsa-miR-25-3p â†' 0.219

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

share_TUM
EDGE FRACTION 0â€“1 72%

in tumour DERIVED

example: CDH1 / hsa-miR-25-3p â†' 0.243

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

SHIFT_ Â· 1

core â€” the columns findings rest on

1 columns Â· rung **edge** Â· coverage **72%â€“72%** (median 72%)
1× categorical

shift_class EDGE CATEGORICAL 72%

Cross-state class for the edge. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' tumour_realized

CAL_ Â· 7

core â€” the columns findings rest on

CALIBRATED posterior width (`learned/calibration.py`): the reported SD corrected for the measured 1.18â€“1.34x narrowness, with `_lo` / `_hi` spanning the correction constant's own +/-1 SD.

7 columns Â· rung **edge** Â· coverage **100%â€“100%** (median 100%) Â· **4** share the block description
4× real â‰¥ 0 Â· 3× boolean

cal_beta_sd EDGE REAL â‰¥ 0 100%

Î² Â· standard deviation DERIVED

example: CDH1 / hsa-miR-25-3p â†' 0.05952

cal_beta_sd_hi EDGE REAL â‰¥ 0 100%

Î² Â· standard deviation DERIVED

example: CDH1 / hsa-miR-25-3p â†' 0.06636

cal_beta_sd_lo EDGE REAL â‰¥ 0 100%

Î² Â· standard deviation DERIVED

example: CDH1 / hsa-miR-25-3p â†' 0.05396

cal_identified EDGE BOOLEAN 100%

Is the edge identified (`|cal_z| > 2`) under the CALIBRATED posterior width â€" the honest version of `identified`: **1,110 vs 1,375 edges**, i.e. **19.9% [18.1â€“22.4]** against 24.8%. AUTHORED

values: False (4,534) Â· True (1,110)

example: CDH1 / hsa-miR-25-3p â†' True

cal_identified_hi EDGE BOOLEAN 100%

The UPPER end of the calibrated identification band â€" `|cal_z| > 2` under the optimistic calibration (990 edges vs 1,110 central). AUTHORED

values: False (4,654) Â· True (990)

example: CDH1 / hsa-miR-25-3p â†' True

cal_identified_lo EDGE BOOLEAN 100%

The LOWER end of the calibrated identification band (1,234 edges vs 1,110 central). See `cal_identified_hi` â€" the pair is the uncertainty, and reporting one alone hides it. AUTHORED

values: False (4,410) Â· True (1,234)

example: CDH1 / hsa-miR-25-3p â†' True

cal_z EDGE REAL â‰¥ 0 100%

z-score DERIVED

example: CDH1 / hsa-miR-25-3p â†' 3.702

ECHIM_ Â· 3

core â€” the columns findings rest on

3 columns Â· rung **edge** Â· coverage **8%â€“26%** (median 22%)
2× count Â· 1× real â‰¥ 0

echim_manakov_w
EDGE | REAL â‰¥ 0 | 22%

Manakov chimeric-eCLIP support weight for this edge (median 3, max 1,712). AUTHORED

example: `ESR1 / hsa-miR-20b-5p â†' 0.3333`

---

echim_n_sources
EDGE | COUNT | 26%

How many chimeric-eCLIP sources support this edge (1 on 1,151 Â· 2 on 262 Â· 3 on 34). AUTHORED

values: `1.0 (1,151) Â· 2.0 (262) Â· 3.0 (34) Â· 4.0 (2)`

example: `CDH1 / hsa-miR-92a-3p â†' 1`

---

echim_tarbase_w
EDGE | COUNT | 8%

TarBase chimeric support weight (median 2, max 28) â€" a much shorter tail than the Manakov twin, and defined on only 8.4% of edges. AUTHORED

example: `CDH1 / hsa-miR-92a-3p â†' 4`

> *Defined on: edges with a chimeric (AGO-ligation) duplex â€" âš per-source, NEVER pooled; cross-tissue, not breast*

RANK_ Â· 4

core â€" the columns findings rest on

> The arm's WITHIN-GENE rank by dose in one state.
>
> **4** columns Â· rung **edge** Â· coverage **46%â€"72%** (median 72%) Â· **4** share the block description
>
> 4× count

rank_HLY
EDGE | COUNT | 72%

in the GTEx-healthy state DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 4`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

---

rank_HLY_meas
EDGE | COUNT | 46%

in the GTEx-healthy state Â· measured only (un-imputed) DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 2`

> *Defined on: edges whose gene has â‰¥1 GTEx-MEASURED regulator â€" âš NaN = GTEx cannot see it (abstention), NOT 0*

---

rank_NAT
EDGE | COUNT | 72%

in matched normal (NAT) DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 2`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

rank_TUM
EDGE | COUNT | 72%

in tumour DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 2`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

CTX_ Â· 18

core â€" the columns findings rest on

> Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
>
> **18** columns Â· **spans arm, gene** â€" this prefix groups columns living on DIFFERENT units Â· coverage **84%â€"100%** (median 99%) Â· **10** share the block description
>
> 4× real â‰¥ 0 Â· 4× signed âˆ'1â€¦1 Â· 3× real, signed Â· 3× count Â· 2× boolean Â· 1× categorical Â· 1× fraction 0â€"1

ctx_apriori_class
GENE | CATEGORICAL | 100%

class label DERIVED

values: `A_COMPETENT (3,989) Â· D_UNDEFINED (654) Â· B_PARTIAL (609) Â· C_WEAK (397)`

example: `CDH1 / hsa-miR-25-3p â†' A_COMPETENT`

---

ctx_arm_abundant
`ARM` `BOOLEAN` `95%`

`ctx_arm_dose >= log2(11)` â€” the arm-abundance gate, ARM-rung on an edge card. See the arm card's entry. `AUTHORED`

values: `True (3,384) Â· False (1,981)`

example: `CDH1 / hsa-miR-25-3p â†’ True`

> *Defined on: arms present on the EDGE card (729 of 2,450) â€” lifted arm-rung columns, names preserved*

---

ctx_arm_dose
`ARM` `REAL â‰¥ 0` `95%`

This arm's median log2(RPM+1) â€” see the arm card's entry. `AUTHORED`

example: `CDH1 / hsa-miR-25-3p â†’ 13.05`

---

ctx_ceiling
`GENE` `SIGNED âˆ’1â€¦1` `99%`

The UPPER BOUND on what ANY estimator can achieve for this gene â€” the 5-fold out-of-fold RÂ² of an UNRESTRICTED OLS on the gene's real design (`card_context.py:15`). `AUTHORED`

example: `CDH1 / hsa-miR-25-3p â†’ 0.1535`

---

ctx_collapse
`GENE` `REAL â‰¥ 0` `100%`

`n_arm / n_fam` for the gene â€” **how much curation BUNDLES same-seed mates onto it** (1â€“5). `AUTHORED`

example: `CDH1 / hsa-miR-25-3p â†’ 1.091`

---

ctx_d_collin
`GENE` `SIGNED âˆ’1â€¦1` `84%`

Within-design COLLINEARITY asymmetry, real minus decoy. The second design-match axis MH-167 tested; also bounded and not moving the gap. `AUTHORED`

example: `CDH1 / hsa-miR-25-3p â†’ -0.02497`

> *Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)*

---

ctx_d_fam
`GENE` `REAL, SIGNED` `99%`

Design-WIDTH asymmetry between the real design and its matched decoy (real minus fake families). `AUTHORED`

example: `CDH1 / hsa-miR-25-3p â†’ 0`

---

ctx_dose_max
`GENE` `REAL â‰¥ 0` `100%`

dose Â· maximum `DERIVED`

example: `CDH1 / hsa-miR-25-3p â†’ 15.38`

---

ctx_dose_med
`GENE` `REAL â‰¥ 0` `100%`

dose Â· median `DERIVED`

example: `CDH1 / hsa-miR-25-3p â†’ 6.318`

---

ctx_frac_abund
`GENE` `FRACTION 0â€“1` `100%`

fraction Â· unweighted abundance-sum reference `DERIVED`

example: `CDH1 / hsa-miR-25-3p â†’ 0.5833`

---

ctx_gap_core
`GENE` `SIGNED âˆ’1â€¦1` `99%`

gap Â· core confounder block `DERIVED`

example: `CDH1 / hsa-miR-25-3p â†’ -0.2714`

---

ctx_gap_d_dose
`GENE` `REAL, SIGNED` `99%`

Dose asymmetry between the real and decoy designs (median âˆ’0.43). `AUTHORED`

example: `CDH1 / hsa-miR-25-3p â†’ 0.1238`

---

ctx_gap_deconv
`GENE` `SIGNED âˆ’1â€¦1` `99%`

gap Â· composition-adjusted (deconvolution block) `DERIVED`

example: `CDH1 / hsa-miR-25-3p â†’ -0.1864`

---

ctx_gap_retention
`GENE` `REAL, SIGNED` `95%`

gap Â· retention `DERIVED`

example: `CDH1 / hsa-miR-25-3p â†’ 0.6867`

---

ctx_measurable
`GENE` `BOOLEAN` `99%`

Is the gene's `ctx_ceiling` above `CEILING_ROBUST` (**0.02**, not 0 â€” MH-144)? `AUTHORED`

values: `True (4,189) Â· False (1,428)`

example: `CDH1 / hsa-miR-25-3p â†’ True`

---

ctx_n_abund
`GENE` `COUNT` `100%`

count Â· unweighted abundance-sum reference `DERIVED`

example: `CDH1 / hsa-miR-25-3p â†’ 7`

---

ctx_n_fam_abund
`GENE` `COUNT` `100%`

count Â· per seed family Â· unweighted abundance-sum reference `DERIVED`

example: `CDH1 / hsa-miR-25-3p â†’ 6`

---

ctx_n_fam_multi

GENE | COUNT | 100%

count Â· per seed family DERIVED

example: CDH1 / hsa-miR-25-3p â†' 1

HEALTHY_ Â· 3

supporting â€" read these when the core raises a question

3 columns Â· rung **arm** Â· identical column set to the arm block Â· coverage **64%â€"72%** (median 64%)
1× categorical Â· 1× real â‰¥ 0 Â· 1× boolean

healthy_leg

ARM | CATEGORICAL | 72%

Provenance of this arm's GTEx healthy leg â€" `measured`, `collapse_blind` or `true_absent`. AUTHORED

values: measured (2,573) Â· collapse_blind (854) Â· true_absent (652)

example: CDH1 / hsa-miR-25-3p â†' measured

healthy_potential

ARM | REAL â‰¥ 0 | 64%

The arm's IMPUTED healthy abundance â€" a repression-POTENTIAL proxy, collapse-repaired (0.27 â€" 17.9). AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 8.33

healthy_uninformative

ARM | BOOLEAN | 64%

Is the healthy leg both `collapse_blind` AND high-potential â€" i.e. the one combination where an *acquired* call is genuinely unsafe? **True on 143 of 3,490 edges (4%).** AUTHORED

values: False (3,490) Â· True (143)

example: CDH1 / hsa-miR-25-3p â†' False

IDENTITY_ Â· 5

supporting â€" read these when the core raises a question

5 columns Â· rung **edge** Â· coverage **96%â€"98%** (median 96%)
2× fraction 0â€"1 Â· 2× real, signed Â· 1× boolean

identity_abs

EDGE | FRACTION 0â€"1 | 96%

|identity| as a fraction of the gene's total |identity|, **clipped to [0,1]**. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.3305

identity_allocated

EDGE | REAL, SIGNED | 96%

Family identity with phantom minor arms forced to 0 â€" the allocation actually used downstream. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.3367

identity_coherence

EDGE | FRACTION 0â€"1 | 96%

`1 / Î£|identity|` over the gene, clipped to (0,1] â€" how much the gene's identity shares CANCEL. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.9817

identity_deconv

EDGE | REAL, SIGNED | 98%

Shapley identity recomputed under the composition-adjusted design. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.2929

identity_reliable

EDGE | BOOLEAN | 96%

Is this edge's Shapley `identity` trustworthy â€" `identity_coherence` â‰¥ 0.5 AND |identity| â‰¤ 1? AUTHORED

values: True (5,272) Â· False (159)

example: CDH1 / hsa-miR-25-3p â†' True

DOSE_ Â· 7

supporting â€” read these when the core raises a question

Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.

7 columns Â· **spans arm, edge** â€" this prefix groups columns living on DIFFERENT units Â· coverage **55%â€"72%** (median 72%) Â· **6** share the block description

3× count Â· 2× real â‰¥ 0 Â· 1× categorical Â· 1× constant

dose_class

EDGE | CATEGORICAL | 72%

class label DERIVED

values: gain (1,620) Â· flat (1,286) Â· loss (1,173)

example: CDH1 / hsa-miR-25-3p â†' flat

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

dose_comp_retention

ARM | REAL â‰¥ 0 | 55%

retention DERIVED

example: CDH1 / hsa-miR-30a-5p â†' 1.02

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

dose_confounded

ARM | CONSTANT | 72%

Is this edge's NATâ†'tumour dose a composition/proliferation artifact (either gated retention < 0.4)? AUTHORED

values: False (4,079)

example: CDH1 / hsa-miR-25-3p â†' False

dose_prolif_retention

ARM | REAL â‰¥ 0 | 55%

retention DERIVED

example: CDH1 / hsa-miR-30a-5p â†' 1.02

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

dose_rank_HLY

EDGE | COUNT | 72%

rank Â· in the GTEx-healthy state DERIVED

example: CDH1 / hsa-miR-25-3p â†' 3

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

dose_rank_NAT

EDGE | COUNT | 72%

rank Â· in matched normal (NAT) DERIVED

example: CDH1 / hsa-miR-25-3p â†' 2

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

dose_rank_TUM

EDGE | COUNT | 72%

rank Â· in tumour DERIVED

example: CDH1 / hsa-miR-25-3p â†' 2

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

D_RANK_ Â· 4

supporting â€" read these when the core raises a question

Change in the arm's WITHIN-GENE dose rank between two states.

4 columns Â· rung **edge** Â· coverage **46%â€"72%** (median 72%) Â· **3** share the block description

4× real, signed

d_rank_HLY_NAT

EDGE | REAL, SIGNED | 72%

in the GTEx-healthy state Â· in matched normal (NAT) DERIVED

example: CDH1 / hsa-miR-25-3p â†' 2

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

d_rank_HLY_TUM

EDGE | REAL, SIGNED | 72%

in the GTEx-healthy state Â· in tumour DERIVED

example: CDH1 / hsa-miR-25-3p â†' 2

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

d_rank_HLY_TUM_meas

EDGE | REAL, SIGNED | 46%

`d_rank_HLY_TUM` recomputed over arms with a MEASURED healthy level only â€” the collapse-safe version. Prefer it over the bare column. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0`

> *Defined on: edges whose gene has â‰¥1 GTEx-MEASURED regulator â€" âš NaN = GTEx cannot see it (abstention), NOT 0*

---

d_rank_NAT_TUM

EDGE | REAL, SIGNED | 72%

in matched normal (NAT) Â· in tumour DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

GENE_ Â· 6

supporting â€" read these when the core raises a question

> Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction.
>
> **6** columns Â· rung **gene** Â· coverage **72%â€"100%** (median 99%) Â· **3** share the block description
> 2× categorical Â· 2× boolean Â· 1× real â‰¥ 0 Â· 1× real, signed

gene_cancer_role

GENE | CATEGORICAL | 100%

â›" **NOT the arm's role â€" the GENE's cancer role** (`oncogene` / `tsg` / `oncogene/tsg` / `unknown`), from `gene_roles.load_gene_roles`. AUTHORED

values: unknown (3,667) Â· oncogene (1,189) Â· tsg (661) Â· oncogene/tsg (132)

example: `CDH1 / hsa-miR-25-3p â†' tsg`

---

gene_dominated

GENE | BOOLEAN | 97%

Does one family carry **more than 60%** of this gene's tumour pressure budget (2,641 of 5,495)? AUTHORED

values: False (2,854) Â· True (2,641)

example: `CDH1 / hsa-miR-25-3p â†' False`

---

gene_iqr

GENE | REAL â‰¥ 0 | 72%

interquartile range DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 1.72`

> *Defined on: genes with a NAT-TUM gene-level leg*

---

gene_lfc_NAT_TUM

GENE | REAL, SIGNED | 72%

log fold-change Â· in matched normal (NAT) Â· in tumour DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0.73`

> *Defined on: genes with a NAT-TUM gene-level leg*

---

gene_net_repr

GENE | BOOLEAN | 100%

Is this gene NET-REPRESSED in tumour by the Â§6b-RETIRED heuristic lane (1,376 of 5,634 edges)? AUTHORED

values: False (4,258) Â· True (1,376)

example: `CDH1 / hsa-miR-25-3p â†' False`

---

gene_repr_class

GENE | CATEGORICAL | 100%

class label DERIVED

values: never_repressed (2,939) Â· lost_repression (1,318) Â· constitutive_repressed (770) Â· gained_repression (606) Â· na (1)

example: `CDH1 / hsa-miR-25-3p â†' never_repressed`

AGO_ Â· 2

supporting â€" read these when the core raises a question

> **2** columns Â· rung **arm** Â· âš ago_loading is rank-identical to the arm card's ago_dom â€" one quantity, two names Â· coverage **95%â€"100%** (median 97%)
> 1× fraction 0â€"1 Â· 1× boolean

ago_loading

ARM | FRACTION 0â€“1 | 95%

AGO/RISC loading discount. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.996

---

ago_loading_measured

ARM | BOOLEAN | 100%

Was `ago_loading` actually MEASURED for this edge (vs defaulted to 1.0). True for 3,820/5,648 edges (67.6%). AUTHORED

values: True (3,820) Â· False (1,829)

example: CDH1 / hsa-miR-25-3p â†’ True

> *Defined on: arms with a Manakov AGO-dominance measurement â€” â›” NOT loading(), which fabricates 1.0*

PIP_ Â· 2

supporting â€” read these when the core raises a question

> **2** columns Â· rung **edge** Â· coverage **100%â€“100%** (median 100%)
> 1× constant Â· 1× fraction 0â€“1

pip_dense

EDGE | CONSTANT | 100%

âš CONSTANT by construction â€” the coupling readout is called with Ï€â‰¡1, so every edge has the same value. AUTHORED

values: 1.0 (5,644)

example: CDH1 / hsa-miR-25-3p â†’ 1

---

pip_discovery

EDGE | FRACTION 0â€“1 | 100%

Posterior inclusion probability. `pip_dense` comes from the pi=1 coupling readout; `pip_discovery` from the evidence-pi readout. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 1

RETENTION_ Â· 3

reference â€” context, provenance and rarely-quoted detail

> **3** columns Â· rung **edge** Â· coverage **72%â€“100%** (median 100%)
> 1× real â‰¥ 0 Â· 1× boolean Â· 1× real, signed

retention_beta

EDGE | REAL â‰¥ 0 | 100%

Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.7885

---

retention_reliable

EDGE | BOOLEAN | 100%

Is this edge's `retention` ratio trustworthy â€” i.e. does it have a real denominator? AUTHORED

values: False (4,269) Â· True (1,375)

example: CDH1 / hsa-miR-25-3p â†’ True

---

retention_rho

EDGE | REAL, SIGNED | 72%

Ï_deconv / Ï_core â€” AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.65

OOF_ Â· 3

reference â€” context, provenance and rarely-quoted detail

> Out-of-fold predictive correlation, arm-level vs family-level, and their difference (`oof_drho` = arm minus family). Negative = the arm-resolved fit predicts repression better.
>
> **3** columns Â· rung **family** Â· coverage **20%â€“20%** (median 20%) Â· **2** share the block description
> 3× signed âˆ’1â€¦1

oof_drho
FAMILY · SIGNED Â^'1Â€¦1 · 20%

Arm-level OOF rho MINUS family-level. AUTHORED
example: CDH1 / hsa-miR-25-3p â†' -0.004735

oof_rho_arm
FAMILY · SIGNED Â^'1Â€¦1 · 20%

Spearman Ï Â· per arm DERIVED
example: CDH1 / hsa-miR-25-3p â†' -0.2411

> *Defined on: multi-arm cells only (n_arm_in_cell > 1) â€" 20.3% of edges*

oof_rho_fam
FAMILY · SIGNED Â^'1Â€¦1 · 20%

Spearman Ï Â· per seed family DERIVED
example: CDH1 / hsa-miR-25-3p â†' -0.2364

> *Defined on: multi-arm cells only (n_arm_in_cell > 1) â€" 20.3% of edges*

GRANK_ Â· 3

reference â€" context, provenance and rarely-quoted detail

> The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.
>
> **3** columns Â· rung **arm** Â· coverage **63%â€"69%** (median 68%) Â· **3** share the block description
> 3× count

grank_HLY
ARM · COUNT · 63%

in the GTEx-healthy state DERIVED
example: CDH1 / hsa-miR-25-3p â†' 96

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

grank_NAT
ARM · COUNT · 68%

in matched normal (NAT) DERIVED
example: CDH1 / hsa-miR-25-3p â†' 99

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

grank_TUM
ARM · COUNT · 69%

in tumour DERIVED
example: CDH1 / hsa-miR-25-3p â†' 99

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

COMPOSITION_ Â· 1

reference â€" context, provenance and rarely-quoted detail

> **1** columns Â· rung **edge** Â· coverage **100%â€"100%** (median 100%)
> 1× categorical

composition_class
EDGE · CATEGORICAL · 100%

Composition-adjustment verdict for this row (e.g. `composition_explained` when the effect does not survive the deconvolution block). AUTHORED
values: cell_intrinsic (3,764) Â· partial (1,167) Â· composition_explained (713)
example: CDH1 / hsa-miR-25-3p â†' cell_intrinsic

W_ Â· 1

reference â€" context, provenance and rarely-quoted detail

> **1** columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)
> 1× real â‰¥ 0

w_max
GENE | REAL Â‰¥ 0 | 100%

Maximum curated EVIDENCE weight over the gene's regulators. Gene-rung repeat. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 13.24`

OWN_ Â· 1

reference â€” context, provenance and rarely-quoted detail

> **1** columns Â· rung **arm** Â· coverage **88%â€“88%** (median 88%)
> 1× real â‰¥ 0

own_specific_frac
ARM | REAL Â‰¥ 0 | 88%

Fraction of the arm's variance that is target-specific. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 0.55`

DGLOBAL_ Â· 3

reference â€” context, provenance and rarely-quoted detail

> Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step.
> **3** columns Â· rung **arm** Â· coverage **63%â€“68%** (median 63%) Â· **3** share the block description
> 3× real, signed

dGlobal_HLY_NAT
ARM | REAL, SIGNED | 63%

in the GTEx-healthy state Â· in matched normal (NAT) DERIVED

example: `CDH1 / hsa-miR-25-3p â†’ 3`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)*

dGlobal_HLY_TUM
ARM | REAL, SIGNED | 63%

in the GTEx-healthy state Â· in tumour DERIVED

example: `CDH1 / hsa-miR-25-3p â†’ 3`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)*

dGlobal_NAT_TUM
ARM | REAL, SIGNED | 68%

in matched normal (NAT) Â· in tumour DERIVED

example: `CDH1 / hsa-miR-25-3p â†’ 0`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

TERM_ Â· 3

reference â€” context, provenance and rarely-quoted detail

> **3** columns Â· rung **edge** Â· coverage **68%â€“68%** (median 68%)
> 2× signed â^’1â€¦1 Â· 1× real, signed

term_ABUND
EDGE | SIGNED Â^’1Â€¦1 | 68%

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ -0.004`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

term_INTERACT
EDGE | SIGNED Â^’1Â€¦1 | 68%

The INTERACTION term â€” the part of the NATâ†’tumour change attributable to abundance and wiring moving TOGETHER. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ -0.003`

term_WIRING
EDGE | REAL, SIGNED | 68%

The WIRING term of the dose-vs-wiring decomposition: how much of the NATâ†’tumour change comes from the arm's M-weight moving rather than its abundance. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ 1.376`

M_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

m_nnls
EDGE | REAL â‰¥ 0 | 100%

Bagged-NNLS weight. Exactly 0 in 31.7% of families â€” this is what lets `identity` say a regulator contributes nothing, which `beta` structurally cannot. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.2293

PRIOR_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **100%â€“100%** (median 100%)
1× constant

prior_pi
EDGE | CONSTANT | 100%

The Gibbs sampler's inclusion prior â€” **CONSTANT 0.05** across the card. AUTHORED

values: 0.05 (5,644)

example: CDH1 / hsa-miR-25-3p â†' 0.05

EDGE_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **73%â€“73%** (median 73%)
1× signed âˆ'1â€¦1

edge_rho_adj
EDGE | SIGNED âˆ'1â€¦1 | 73%

An edge-rung quantity carried onto this row. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.198

> *Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)*

MEAN_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **90%â€“90%** (median 90%)
2× real, signed

mean_dGlobalRank
ARM | REAL, SIGNED | 90%

Mean change in the arm's GLOBAL abundance percentile across the paired patients. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' 0.01

mean_own_shift
ARM | REAL, SIGNED | 90%

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises. AUTHORED

example: CDH1 / hsa-miR-25-3p â†' -0.194

> *Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)*

SURROGATE_ Â· 2

reference â€” context, provenance and rarely-quoted detail

> **2** columns Â· rung **arm** Â· coverage **3%â€“3%** (median 3%)
> 1× fraction 0â€“1 Â· 1× categorical

surrogate_corr

ARM | FRACTION 0â€“1 | 3%

How well the surrogate tracks the arm where both are observed (0.54 â€“ 0.89). AUTHORED

values: 0.874 (43) Â· 0.892 (35) Â· 0.621 (29) Â· 0.842 (26) Â· 0.541 (23) Â· 0.606 (18) Â· 0.728 (16)

example: CDH1 / hsa-miR-30a-5p â†’ 0.606

surrogate_instrument

ARM | CATEGORICAL | 3%

Which SURROGATE stands in for a collapsed GTEx healthy leg (7 levels; defined on 3% of edges). AUTHORED

values: hsa-miR-20a-5p (43) Â· hsa-miR-181b-5p (35) Â· hsa-miR-93-5p (29) Â· hsa-miR-429 (26) Â· hsa-let-7b-5p (23) Â· hsa-miR-30c-5p (18) Â· hsa-miR-221-3p (16)

example: CDH1 / hsa-miR-30a-5p â†’ hsa-miR-30c-5p

ESUB_ Â· 8

reference â€” context, provenance and rarely-quoted detail

> PAM50 subtype-stratified coupling at the EDGE rung â€” one fit per subtype plus the pooled fit and an adjusted q.
> **8** columns Â· rung **edge** Â· coverage **98%â€“100%** (median 99%) Â· **5** share the block description
> 6× signed âˆ’1â€¦1 Â· 1× categorical Â· 1× p-value

esub_best_sub

EDGE | CATEGORICAL | 100%

the best DERIVED

values: Her2 (1,782) Â· Basal (1,679) Â· LumB (1,178) Â· LumA (1,005)

example: CDH1 / hsa-miR-25-3p â†’ Her2

> *Defined on: edges with a per-PAM50 heterogeneity fit â€” âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_q_adj

EDGE | P-VALUE | 98%

BH-adjusted q for the best-subtype contrast â€” AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.3849

esub_rho_Basal

EDGE | SIGNED âˆ’1â€¦1 | 99%

Spearman Ï DERIVED

example: CDH1 / hsa-miR-25-3p â†’ -0.1683

> *Defined on: edges with a per-PAM50 heterogeneity fit â€” âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_rho_Her2

EDGE | SIGNED âˆ’1â€¦1 | 98%

Spearman Ï DERIVED

example: CDH1 / hsa-miR-25-3p â†’ -0.3531

> *Defined on: edges with a per-PAM50 heterogeneity fit â€” âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_rho_LumA

EDGE | SIGNED âˆ’1â€¦1 | 99%

Spearman Ï DERIVED

example: CDH1 / hsa-miR-25-3p â†’ -0.174

> *Defined on: edges with a per-PAM50 heterogeneity fit â€” âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_rho_LumB

EDGE | SIGNED âˆ’1â€¦1 | 99%

Spearman Ï DERIVED

example: CDH1 / hsa-miR-25-3p â†’ -0.1127

> *Defined on: edges with a per-PAM50 heterogeneity fit â€” âš DISTRIBUTIONAL (MH-165), not a per-edge subtype label*

esub_rho_best

EDGE | SIGNED Âˆ'1Â€¦1 | 100%

Coupling in the arm's BEST PAM50 subtype. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.3531`

esub_rho_pool

EDGE | SIGNED Âˆ'1Â€¦1 | 100%

Coupling pooled across subtypes â€” the unselected reference for `esub_rho_best` (median âˆ'0.0135, indistinguishable from a random-subtype draw at âˆ'0.0123). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.1909`

REALIZATION_ Â· 2

reference â€” context, provenance and rarely-quoted detail

> **2** columns Â· rung **edge** Â· coverage **58%â€"69%** (median 63%)
> 1× fraction 0â€"1 Â· 1× real, signed

realization_identity

EDGE | FRACTION 0Â€"1 | 58%

Within-patient realization quantities â€” the score, its identity allocation, and who owns it. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.849`

> *Defined on: genes with a family-level realization fit*

realization_score

EDGE | REAL, SIGNED | 69%

`âˆ'coupling_z_tum` â€” the calibrated tumour repression strength as a positive score, so bigger = more repressed. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 3.04`

KD_ Â· 2

reference â€” context, provenance and rarely-quoted detail

> **2** columns Â· rung **edge** Â· coverage **85%â€"85%** (median 85%)
> 1× percent Â· 1× real, signed

kd_affinity_pct

EDGE | PERCENT | 85%

Is this gene among the arm's strongest predicted targets â€” a WITHIN-ARM percentile, **higher = stronger target**. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.8387`

kd_repression

EDGE | REAL, SIGNED | 85%

scanMiR RBNS binding affinity. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.8171`

WIRING_ Â· 1

reference â€” context, provenance and rarely-quoted detail

> **1** columns Â· rung **edge** Â· coverage **68%â€"68%** (median 68%)
> 1× fraction 0â€"1

wiring_frac

EDGE | FRACTION 0Â€"1 | 68%

Fraction of the acquisition score attributable to WIRING rather than abundance. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 1`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

CELL_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **20%â€“20%** (median 20%)
1× boolean

cell_arms_resolvable
FAMILY BOOLEAN 20%

Are the cell's arms separable AND does splitting them help out-of-fold. True for 31.7% of edges in multi-arm cells. AUTHORED

values: False (783) Â· True (364)

example: CDH1 / hsa-miR-25-3p â†’ True

SD_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **20%â€“20%** (median 20%)
1× real â‰¥ 0

sd_arm
EDGE REAL â‰¥ 0 20%

Posterior SD of the arm-level Î². AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 0.01097

ACQUISITION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **59%â€“59%** (median 59%)
1× real, signed

acquisition_score
EDGE REAL, SIGNED 59%

Composite acquisition score across states. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 3.35

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)*

Z_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **20%â€“20%** (median 20%)
1× real â‰¥ 0

z_arm
EDGE REAL â‰¥ 0 20%

`beta_arm / sd_arm` from the WITHIN-CELL fit (`arm_rung.py`) â€” the arm's z inside its (gene, seed_family) cell only, defined on the 20.3% of edges in multi-arm cells. AUTHORED

example: CDH1 / hsa-miR-25-3p â†’ 3.706

DSHARE_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **edge** Â· coverage **65%â€“65%** (median 65%)
2× signed âˆ’1â€¦1

dShare_M_own
EDGE | SIGNED Â˜'1Â€¦1 | 65%

Mean over the 103 paired patients of this arm's change in M-WEIGHTED share of its gene's total regulator dose, tumour minus that patient's OWN NAT. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.045`

---

dShare_raw_own
EDGE | SIGNED Â˜'1Â€¦1 | 65%

As `dShare_M_own` but on RAW abundance share, with the model's `M` weights removed. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.04`

HEUR_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)
1× count

heur_gene_nreg
GENE | COUNT | 100%

Regulator count for this gene from the Â§6b-RETIRED heuristic lane. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 12`

CPTAC_ Â· 20

reference â€" context, provenance and rarely-quoted detail

CPTAC protein channel.

20 columns Â· rung **edge** Â· per-edge; the gene card aggregates it Â· coverage **45%â€"79%** (median 65%) Â· **12** share the block description
12× signed â˜'1â€¦1 Â· 4× real, signed Â· 2× constant Â· 2× categorical

cptac_prosp_comp_class_prot
EDGE | CATEGORICAL | 59%

Composition class of this edge's PROTEIN coupling (cell_intrinsic / partial / composition_explained) in the prospective cohort. AUTHORED

values: `composition_explained (1,427) Â· cell_intrinsic (1,278) Â· partial (647)`

example: `CDH1 / hsa-miR-25-3p â†' composition_explained`

---

cptac_prosp_n
EDGE | CONSTANT | 79%

Patients in the prospective CPTAC cohort â€" **CONSTANT 101**. AUTHORED

values: `101.0 (4,486)`

example: `CDH1 / hsa-miR-25-3p â†' 101`

---

cptac_prosp_ret_prot
EDGE | REAL, SIGNED | 59%

RETENTION at EDGE rung â€" this edge's composition-ADJUSTED protein coupling divided by its UNADJUSTED one, in the CPTAC prospective cohort. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.2189`

---

cptac_prosp_ret_rna
EDGE | REAL, SIGNED | 61%

As `cptac_prosp_ret_prot` but against **mRNA** rather than protein: adjusted Ã· unadjusted mRNA coupling for this edge, prospective cohort. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.3`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_disc
EDGE | SIGNED Â˜'1Â€¦1 | 78%

CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against the discovery layer DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0.08776`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_disc_raw
EDGE | SIGNED Â˜'1Â€¦1 | 78%

CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against the discovery layer Â· UNADJUSTED â€" no confounder block DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0.014`

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_prosp_rho_prot

EDGE | SIGNED Âˆ'1Â€¦1 | 78%

CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against protein DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0.03748`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_prot_raw

EDGE | SIGNED Âˆ'1Â€¦1 | 79%

UNADJUSTED protein coupling, prospective. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.1712`

---

cptac_prosp_rho_rna

EDGE | SIGNED Âˆ'1Â€¦1 | 79%

CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against mRNA DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0.06188`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_prosp_rho_rna_raw

EDGE | SIGNED Âˆ'1Â€¦1 | 79%

CPTAC prospective cohort (nâ‰ˆ101) Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€" no confounder block DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0.2063`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_comp_class_prot

EDGE | CATEGORICAL | 45%

CPTAC TCGA-overlap cohort (n=105) Â· class label Â· against protein DERIVED

values: `cell_intrinsic (1,447) Â· composition_explained (632) Â· partial (463)`

example: `CDH1 / hsa-miR-25-3p â†' composition_explained`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_n

EDGE | CONSTANT | 70%

Patients in the TCGA-overlap CPTAC cohort â€" **CONSTANT 105**. See the prospective twin. AUTHORED

values: `105.0 (3,968)`

example: `CDH1 / hsa-miR-25-3p â†' 105`

---

cptac_t105_ret_prot

EDGE | REAL, SIGNED | 45%

Retention at edge rung on the TCGA-overlap cohort â€" see the prospective twin. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' -0.5722`

---

cptac_t105_ret_rna

EDGE | REAL, SIGNED | 49%

As the prospective mRNA twin, on the TCGA-overlap cohort. AUTHORED

example: `CDH1 / hsa-miR-25-3p â†' 0.6976`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_disc

EDGE | SIGNED Âˆ'1Â€¦1 | 64%

CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against the discovery layer DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0.05624`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_disc_raw

EDGE | SIGNED Âˆ'1Â€¦1 | 65%

CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against the discovery layer Â· UNADJUSTED â€" no confounder block DERIVED

example: `CDH1 / hsa-miR-25-3p â†' -0.03572`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_prot

EDGE | SIGNED Âˆ'1Â€¦1 | 64%

CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against protein DERIVED

example: `CDH1 / hsa-miR-25-3p â†' 0.0313`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_prot_raw

EDGE | SIGNED Âˆ'1Â€¦1 | 65%

CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against protein Â· UNADJUSTED â€" no confounder block DERIVED

example: `CDH1 / hsa-miR-25-3p â†' -0.0547`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_rho_rna

EDGE | SIGNED Âˆ'1Â€¦1 | 64%

CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against mRNA DERIVED

example: `CDH1 / hsa-miR-25-3p â†' -0.06411`

> *Defined on: genes/edges present in the CPTAC cohort*

cptac_t105_rho_rna_raw
EDGE | SIGNED âˆ’1â€¦1 | 65%

CPTAC TCGA-overlap cohort (n=105) Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€” no confounder block DERIVED

example: `CDH1 / hsa-miR-25-3p â†’ -0.0919`

> *Defined on: genes/edges present in the CPTAC cohort*

FAMILY_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **69%â€“69%** (median 69%)
1× signed âˆ’1â€¦1

family_rho_adj
FAMILY | SIGNED âˆ’1â€¦1 | 69%

Family-level properties carried onto this row (size, the arm's dose share of it, and its role). AUTHORED

example: `CDH1 / hsa-miR-25-3p â†’ -0.332`

HLY_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **61%â€“61%** (median 61%)
1× boolean

hly_leg_concordant
EDGE | BOOLEAN | 61%

Do the two HLYâ†’TUM legs (`_QN` and `_raw`) at least AGREE ON SIGN. **Concordant 76.1%, DISCORDANT 23.9%** of the 3,433 edges carrying both. AUTHORED

values: `True (2,611) Â· False (822)`

example: `CDH1 / hsa-miR-25-3p â†’ True`

IS_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **edge** Â· coverage **58%â€“58%** (median 58%)
1× boolean

is_realization_owner
EDGE | BOOLEAN | 58%

A boolean flag; read the suffix. AUTHORED

values: `False (2,498) Â· True (759)`

example: `CDH1 / hsa-miR-25-3p â†’ True`

> *Defined on: genes with a family-level realization fit*

realization/gene_card.tsv

gene

One row per the gene's total incoming regulation. One row per gene, so nothing here is checkable by invariance â€” `agg_of` carries the meaning instead.

135 columns **1,549** rows **29** blocks median coverage **83%**

Contents â€” 29 blocks, most important first

N_ Â· 7

core â€” the columns findings rest on

7 columns Â· rung **gene** Â· coverage **91%â€“100%** (median 100%)
7× count

n_arms
GENE COUNT AGG ARM 100%

Arms in the LEARNED MODEL's design â€” `X.shape[1]` from `assemble_gene`, carried via `gene_atlas`. **This is the width the Gibbs actually fit** (median 2, max 90). AUTHORED
example: CDH1 â†' 26

n_cell_intrinsic
GENE COUNT 100%

FAMILIES whose coupling SURVIVES the composition block (retention â‰¥ 0.7). AUTHORED
example: CDH1 â†' 22

n_composition_explained
GENE COUNT 100%

FAMILIES whose coupling does NOT survive the composition block (retention < 0.4). AUTHORED
values: 0 (1,133) Â· 1 (286) Â· 2 (80) Â· 3 (26) Â· 4 (13) Â· 6 (7) Â· 5 (2) Â· 8 (2)
example: CDH1 â†' 0

n_fam
GENE COUNT AGG FAMILY 100%

How many seed families regulate this gene. AUTHORED
example: CDH1 â†' 25

n_hallmark_sets
GENE COUNT 91%

How many MSigDB Hallmark programs contain this gene (1â€“10, median 2). Membership only â€” it says nothing about regulation. AUTHORED
example: CDH1 â†' 3

n_identified
GENE · COUNT · 100%

Families reaching |z| > 2 on the UNCALIBRATED SD. **Zero for 691 genes (44.6%)** â€" meaning the model cannot distinguish ANY of that gene's families from zero. AUTHORED
example: `CDH1 â†' 4`

---

n_pip_disc_gt50
GENE · COUNT · 100%

Count of the gene's families with `pip_discovery > 0.5` under the evidence-Ï€ readout. AUTHORED
example: `CDH1 â†' 4`

(BARE) Â· 2

core â€" the columns findings rest on

---

2 columns Â· **spans gene, key** â€" this prefix groups columns living on DIFFERENT units Â· coverage **100%â€"100%** (median 100%)
1× fraction 0â€"1 Â· 1× identifier

concentration
GENE · FRACTION 0â€"1 · 100%

`beta.max() / beta.sum()` â€" the TOP FAMILY's share of the gene's total Î²; median **0.954**, i.e. most genes are dominated by one family. AUTHORED
example: `CDH1 â†' 0.191`

---

gene
KEY · IDENTIFIER · 100%

KEY â€" HGNC gene symbol. AUTHORED
example: `JUN`

COMP_ Â· 16

core â€" the columns findings rest on

Cell-composition drivers of this gene's coupling (`compartment_*`): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.

16 columns Â· rung **gene** Â· coverage **59%â€"90%** (median 73%) Â· **2** share the block description
6× categorical Â· 6× signed â^'1â€¦1 Â· 3× real, signed Â· 1× real â‰¥ 0

comp_cptac_prot_driver
GENE · CATEGORICAL · 59%

WHICH deconvolved compartment most explains this gene's PROTEIN coupling â€" `prolif` 219 Â· `CAFs` 173 Â· `purity` 140 Â· `PVL` 77 Â· `Endothelial` 67. AUTHORED
example: `CDH1 â†' T-cells`

---

comp_cptac_prot_driver_ret
GENE · REAL, SIGNED · 59%

`1 â^' comp_cptac_prot_driver_share`, exactly. Median +0.398 raw, **+0.469** gated to |Ï_raw| â‰¥ 0.10 â€" quote the gated value. See the mRNA twin. AUTHORED
example: `CDH1 â†' 0.5085`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_driver_share
GENE · REAL, SIGNED · 59%

Protein-layer twin of `comp_tcga_mrna_driver_share`. Same exact complementarity with `_driver_ret` (8.9e-16 on 918 rows). AUTHORED
example: `CDH1 â†' 0.4915`

---

comp_cptac_prot_rho_raw
GENE · SIGNED â^'1â€¦1 · 73%

Unadjusted aggregate PROTEIN coupling â€" denominator of `comp_cptac_prot_driver_ret`. AUTHORED
example: `CDH1 â†' -0.1416`

---

comp_cptac_prot_top_budget
GENE · CATEGORICAL · 73%

CPTAC Â· against protein Â· the top-ranked Â· pressure budget DERIVED
example: `CDH1 â†' Normal Epithelial`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_top_budget_r

`GENE` `SIGNED Â^'1Â€¦1` `73%`

As the TCGA mRNA twin, on CPTAC protein â€” the budget-vs-compartment correlation. Median **â^'0.118**, i.e. it runs the OPPOSITE way to mRNA (+0.121). `AUTHORED`

example: `CDH1 â†' 0.3126`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_top_target

`GENE` `CATEGORICAL` `73%`

CPTAC Â· against protein Â· the top-ranked Â· the target gene's expression `DERIVED`

example: `CDH1 â†' T-cells`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_cptac_prot_top_target_r

`GENE` `SIGNED Â^'1Â€¦1` `73%`

As the TCGA mRNA twin, on CPTAC protein â€” the target-vs-compartment correlation. Median **+0.203**. `AUTHORED`

example: `CDH1 â†' -0.2385`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_tcga_mrna_driver

`GENE` `CATEGORICAL` `66%`

WHICH compartment drives the gene's mRNA coupling â€” the one whose removal moves it most (`CAFs` 279 Â· `purity` 174 Â· `prolif` 148 Â· `Endothelial` 122 Â· `Myeloid` 109). `AUTHORED`

example: `CDH1 â†' T-cells`

---

comp_tcga_mrna_driver_ret

`GENE` `REAL, SIGNED` `66%`

`1 â^' comp_tcga_mrna_driver_share`, exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. `AUTHORED`

example: `CDH1 â†' 0.8681`

---

comp_tcga_mrna_driver_share

`GENE` `REAL Â%¥ 0` `66%`

Share of the gene's mRNA coupling attributable to the single compartment that most changes it â€” **axiom 8's composition GATE**, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. `AUTHORED`

example: `CDH1 â†' 0.1319`

---

comp_tcga_mrna_rho_raw

`GENE` `SIGNED Â^'1Â€¦1` `90%`

The gene's UNADJUSTED aggregate mRNA coupling â€” the DENOMINATOR of `comp_tcga_mrna_driver_ret`. `AUTHORED`

example: `CDH1 â†' -0.1921`

---

comp_tcga_mrna_top_budget

`GENE` `CATEGORICAL` `90%`

The compartment the gene's miRNA BUDGET loads on most â€” a property of the REGULATORS. `AUTHORED`

example: `CDH1 â†' Normal Epithelial`

---

comp_tcga_mrna_top_budget_r

`GENE` `SIGNED Â^'1Â€¦1` `90%`

Correlation between the gene's **miRNA PRESSURE BUDGET** and the fraction of the compartment that budget loads on most (`comp_tcga_mrna_top_budget`), in TCGA mRNA. `AUTHORED`

example: `CDH1 â†' 0.2545`

---

comp_tcga_mrna_top_target

`GENE` `CATEGORICAL` `90%`

The compartment the TARGET GENE's expression loads on most. `AUTHORED`

example: `CDH1 â†' prolif`

---

comp_tcga_mrna_top_target_r

`GENE` `SIGNED Â^'1Â€¦1` `90%`

Correlation between the **TARGET GENE's own expression** and the fraction of the compartment it loads on most. `AUTHORED`

example: `CDH1 â†' 0.3289`

CTX_ Â· 16

core â€” the columns findings rest on

Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

16 columns Â· rung **gene** Â· SUBSET of the edge block, identical rungs Â· coverage **45%â€“100%** (median 90%) Â· **11** share the block description

4× signed â^'1â€¦1 Â· 3× real â‰¥ 0 Â· 3× real, signed Â· 3× count Â· 1× categorical Â· 1× fraction 0â€“1 Â· 1× boolean

ctx_apriori_class
GENE | CATEGORICAL | 100%

A-priori competence class from the gene atlas (A_COMPETENT / B_PARTIAL / C_WEAK / D_UNDEFINED). AUTHORED

values: D_UNDEFINED (730) Â· A_COMPETENT (428) Â· B_PARTIAL (295) Â· C_WEAK (96)

example: CDH1 â†' A_COMPETENT

---

ctx_ceiling
GENE | SIGNED Â^'1â€¦1 | 98%

The UPPER BOUND on what ANY estimator can achieve for this gene â€” the 5-fold out-of-fold RÂ² of an UNRESTRICTED OLS on the gene's real design (`card_context.py:15`). AUTHORED

example: CDH1 â†' 0.1535

---

ctx_collapse
GENE | REAL Â‰¥ 0 | 90%

`n_arm / n_fam` for the gene â€” **how much curation BUNDLES same-seed mates onto it** (1â€“5). AUTHORED

example: CDH1 â†' 1.091

---

ctx_d_collin
GENE | SIGNED Â^'1â€¦1 | 45%

change Â· collinearity DERIVED

example: CDH1 â†' -0.02497

> *Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)*

---

ctx_d_fam
GENE | REAL, SIGNED | 87%

change Â· per seed family DERIVED

example: CDH1 â†' 0

---

ctx_dose_max
GENE | REAL Â‰¥ 0 | 90%

dose Â· maximum DERIVED

example: CDH1 â†' 15.38

---

ctx_dose_med
GENE | REAL Â‰¥ 0 | 90%

dose Â· median DERIVED

example: CDH1 â†' 6.318

---

ctx_frac_abund
GENE | FRACTION 0Â€“1 | 90%

fraction Â· unweighted abundance-sum reference DERIVED

example: CDH1 â†' 0.5833

---

ctx_gap_core
GENE | SIGNED Â^'1Â€¦1 | 87%

Real-minus-decoy coupling gap for this gene. Pooled it is ~-0.012, but the per-gene IQR STRADDLES ZERO â€” this is a set-level distributional shift, not a per-gene effect. Do not rank genes by it. AUTHORED

example: CDH1 â†' -0.2714

---

ctx_gap_d_dose
GENE | REAL, SIGNED | 87%

gap Â· change Â· dose DERIVED

example: CDH1 â†' 0.1238

---

ctx_gap_deconv
GENE | SIGNED Â^'1Â€¦1 | 87%

gap Â· composition-adjusted (deconvolution block) DERIVED

example: CDH1 â†' -0.1864

---

ctx_gap_retention
GENE | REAL, SIGNED | 83%

gap Â· retention DERIVED

example: CDH1 â†' 0.6867

---

ctx_measurable
GENE | BOOLEAN | 98%

Is the gene's `ctx_ceiling` above `CEILING_ROBUST` (**0.02**, not 0 â€” MH-144)? AUTHORED

values: False (845) Â· True (672)

example: CDH1 â†' True

---

ctx_n_abund
GENE | COUNT | 90%

count Â· unweighted abundance-sum reference DERIVED
example: CDH1 â†’ 7

---

ctx_n_fam_abund
GENE | COUNT | 90%

count Â· per seed family Â· unweighted abundance-sum reference DERIVED
example: CDH1 â†’ 6

---

ctx_n_fam_multi
GENE | COUNT | 90%

count Â· per seed family DERIVED
example: CDH1 â†’ 1

TOP_ Â· 8

supporting â€” read these when the core raises a question

> The gene's leading regulator and its value, by magnitude or by identity.
>
> **8** columns Â· rung **gene** Â· coverage **78%â€“100%** (median 89%) Â· **3** share the block description
> 3× identifier Â· 2× real â‰¥ 0 Â· 1× fraction 0â€“1 Â· 1× signed âˆ’1â€¦1 Â· 1× count

top_beta
GENE | REAL â‰¥ 0
AGG FAMILY | 100%

Î² DERIVED
example: CDH1 â†’ 0.2584

---

top_beta_frac
GENE | FRACTION 0â€“1
AGG FAMILY | 100%

Î² Â· fraction DERIVED
example: CDH1 â†’ 0.1919

---

top_family_identity
GENE | IDENTIFIER
AGG FAMILY | 87%

The family Shapley/LMG credits with the gene's regulation. NaN for 197 genes (12.7%) where NNLS zeroed everything â€” an honest 'undefined', which the magnitude answer cannot express. AUTHORED
example: CDH1 â†’ miR-25-3p/32-5p/92-3p/363-3p/367-3p

---

top_family_magnitude
GENE | IDENTIFIER
AGG FAMILY | 100%

The family with the largest beta. AUTHORED
example: CDH1 â†’ miR-25-3p/32-5p/92-3p/363-3p/367-3p

---

top_gaining_arms
GENE | IDENTIFIER | 91%

arms DERIVED
example: CDH1 â†’ hsa-miR-30a-5p(+0.25);hsa-miR-9-5p(+0.24);hsa-miR-544a-3p(+0.02);hsa-miR-34c-3p(+0.01);hsa-miR-25-3p(+0.01);hsa-miR-199a-5p(-0.00)

---

top_identity
GENE | REAL â‰¥ 0
AGG FAMILY | 87%

The largest `identity` share among the gene's families. AUTHORED
example: CDH1 â†’ 0.4477

---

top_identity_gated
GENE | SIGNED âˆ’1â€¦1 | 78%

`top_identity` recomputed over `identity_reliable` edges only â€” **max 1.0000, 0 genes above 1** (vs +740.007 ungated). Defined on 1,203 genes. AUTHORED
example: CDH1 â†’ 0.3367

---

top_identity_n_reliable
GENE | COUNT | 78%

How many of the gene's families survive the identity gate â€” the DENOMINATOR behind `top_identity_gated`. **Median 2.** AUTHORED
example: CDH1 â†’ 26

REALIZED_ Â· 5

supporting â€” read these when the core raises a question

> Realized (as opposed to potential) coupling for the gene â€” the raw and adjusted rho, its retention, and how many regulators it was computed over.
>
> **5** columns Â· rung **gene** Â· coverage **63%â€“82%** (median 81%) Â· **3** share the block description
> 2× signed âˆ’1â€¦1 Â· 1× boolean Â· 1× count Â· 1× real, signed

realized_composition_explained
`GENE` `BOOLEAN` `82%`

Is the gene's REALIZED coupling composition-explained (retention < 0.4)? `AUTHORED`
values: `False (785) Â· True (490)`
example: `CDH1 â†’ True`

---

realized_n_reg
`GENE` `COUNT` `82%`

count `DERIVED`
example: `CDH1 â†’ 17`

---

realized_retention
`GENE` `REAL, SIGNED` `63%`

Realized-coupling retention, Ï_adjusted / Ï_raw. `AUTHORED`
example: `CDH1 â†’ -0.16`

---

realized_rho_adj
`GENE` `SIGNED Âˆ’1Â€¦1` `81%`

Spearman Ï Â· composition-adjusted `DERIVED`
example: `CDH1 â†’ -0.035`

---

realized_rho_raw
`GENE` `SIGNED Âˆ’1Â€¦1` `81%`

Spearman Ï Â· UNADJUSTED â€” no confounder block `DERIVED`
example: `CDH1 â†’ 0.225`

W_ Â· 1

supporting â€” read these when the core raises a question

> **1** columns Â· rung **gene** Â· coverage **100%â€“100%** (median 100%)
> 1× real â‰¥ 0

w_max
`GENE` `REAL Â‰¥ 0` `100%`

The MAXIMUM curated evidence weight `w` over the gene's regulators (`gene_atlas` : `nanmax(w)`), where `w` comes from the same PMID-deduped ledger that feeds `lit_` and `fame_` . `AUTHORED`
example: `CDH1 â†’ 13.24`

LIT_ Â· 7

supporting â€” read these when the core raises a question

> **7** columns Â· rung **gene** Â· coverage **21%â€“21%** (median 21%)
> 4× count Â· 2× boolean Â· 1× identifier

lit_agrees_identity
`GENE` `BOOLEAN` `21%`

Does `identity` name the same family as the **most-published** one for this gene â€” **True on 100 of 323 (31%)**. `AUTHORED`
values: `False (223) Â· True (100)`
example: `CDH1 â†’ False`

---

lit_agrees_magnitude
`GENE` `BOOLEAN` `21%`

Does the model's largest-Î² family match the most-published one (**96 of 329 genes, 29%**)? `AUTHORED`
values: `False (233) Â· True (96)`
example: `CDH1 â†’ False`

lit_family
GENE · IDENTIFIER · 21%

The most-published seed family for this gene by distinct low-throughput functional PMIDs. AUTHORED

example: CDH1 â†’ miR-9-5p

---

lit_margin
GENE · COUNT · 21%

Gap between the top and second literature family, in PMIDs â€” **median 1.00**, i.e. the literature's 'winner' is usually decided by a SINGLE paper. â‡’ **gate on this before reading `lit_agrees_*`**: a 1-PMID margin is a coin flip dressed as ground truth. AUTHORED

example: CDH1 â†’ 3

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

---

lit_n_lit_families
GENE · COUNT · 21%

â›”â›” **A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH â€" read the name with suspicion.** `eval/lit_ground_truth.py` defines the 'canonical' family of a gene as the **argmax of DISTINCT PubMed IDs** carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded). AUTHORED

example: CDH1 â†’ 25

---

lit_n_pmid
GENE · COUNT · 21%

Distinct low-throughput functional PMIDs behind `lit_family`. Median **3** â€" the 'ground truth' rests on a handful of papers per gene. AUTHORED

example: CDH1 â†’ 6

> *Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)*

---

lit_tier
GENE · COUNT · 21%

Literature-evidence tier for the gene (levels 2, 3 and 4; 4 is the most populated at 132 genes). AUTHORED

values: 4.0 (132) Â· 2.0 (129) Â· 3.0 (68)

example: CDH1 â†’ 4

FRAC_ Â· 3

reference â€" context, provenance and rarely-quoted detail

> **3** columns Â· rung **gene** Â· coverage **60%â€"100%** (median 82%)
> 3× fraction 0â€"1

frac_edges_realized
GENE · FRACTION 0â€"1 · 82%

Fraction of the gene's scored edges that realize a coupling below the threshold. AUTHORED

example: CDH1 â†’ 0.7059

---

frac_gain_true_acquired
GENE · FRACTION 0â€"1 · 60%

Fraction of the gene's acquired-repression calls that survive the healthy-leg triage â€" i.e. are not `collapse_blind` artifacts. AUTHORED

example: CDH1 â†’ 0

---

frac_identified
GENE · FRACTION 0â€"1 · 100%

Fraction of this gene's families reaching `|z| > 2` on the UNCALIBRATED SD. AUTHORED

example: CDH1 â†’ 0.16

CPTAC_ Â· 24

reference â€" context, provenance and rarely-quoted detail

> CPTAC protein channel.
>
> **24** columns Â· rung **gene** Â· AGGREGATE of the edge block via an agg_/abund_ infix â€" NOT derivable (MH-262) Â· coverage **42%â€"89%** (median 66%) Â· **18** share the block description
> 18× signed âˆ'1â€¦1 Â· 2× boolean Â· 2× real, signed Â· 2× count

`cptac_prosp_abund_rho_disc`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `72%`

CPTAC prospective cohort (nâ‰ˆ101) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against the discovery layer `DERIVED`

example: `CDH1 â†' -0.08033`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_abund_rho_prot`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `73%`

CPTAC prospective cohort (nâ‰ˆ101) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against protein `DERIVED`

example: `CDH1 â†' 0.109`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_abund_rho_rna`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `74%`

CPTAC prospective cohort (nâ‰ˆ101) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against mRNA `DERIVED`

example: `CDH1 â†' 0.2848`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_beats_abund_prot`
`GENE` `BOOLEAN` `73%`

Does the learned Î²-weighted aggregate couple more negatively to PROTEIN than a plain abundance sum, in the CPTAC prospective cohort? `AUTHORED`

values: `False (770) Â· True (357)`

example: `CDH1 â†' True`

`cptac_prosp_agg_ret_prot`
`GENE` `REAL, SIGNED` `59%`

`rho_prot(composition-adjusted) / rho_prot_raw` â€" the ADJUSTMENT sense of retention, on the prospective cohort. âœ… **The denominator is ALREADY GATED** at `RHO_GATE` in `cptac_card.py:322`, so values outside [0,1] are NOT the vanishing-denominator artifact of axiom 5 (verified: 0 rows with |raw| < 0.02). `AUTHORED`

example: `CDH1 â†' 0.855`

`cptac_prosp_agg_rho_disc`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `72%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against the discovery layer `DERIVED`

example: `CDH1 â†' -0.1539`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_disc_raw`
`GENE` `SIGNED Â^'1Â€¦1` `72%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against the discovery layer Â· UNADJUSTED â€" no confounder block `DERIVED`

example: `CDH1 â†' -0.1971`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_prot`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `73%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against protein `DERIVED`

example: `CDH1 â†' -0.1211`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_prot_raw`
`GENE` `SIGNED Â^'1Â€¦1` `73%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against protein Â· UNADJUSTED â€" no confounder block `DERIVED`

example: `CDH1 â†' -0.1416`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_rna`
`GENE` `SIGNED Â^'1Â€¦1` `AGG ARM` `74%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against mRNA `DERIVED`

example: `CDH1 â†' 0.04264`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_agg_rho_rna_raw`
`GENE` `SIGNED Â^'1Â€¦1` `74%`

CPTAC prospective cohort (nâ‰ˆ101) Â· Î²-weighted aggregate Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€" no confounder block `DERIVED`

example: `CDH1 â†' 0.06539`

> *Defined on: genes/edges present in the CPTAC cohort*

`cptac_prosp_n_arms`
`GENE` `COUNT` `89%`

Arms contributing to this gene's prospective-cohort aggregate (median 2, max 61). `AUTHORED`

example: `CDH1 â†' 23`

---

cptac_t105_abund_rho_disc
GENE | SIGNED Âˆ’1Â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against the discovery layer DERIVED

example: `CDH1 â†’ -0.0005806`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_abund_rho_prot
GENE | SIGNED Âˆ’1Â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against protein DERIVED

example: `CDH1 â†’ 0.01071`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_abund_rho_rna
GENE | SIGNED Âˆ’1Â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· unweighted abundance-sum reference Â· Spearman Ï Â· against mRNA DERIVED

example: `CDH1 â†’ 0.06946`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_agg_beats_abund_prot
GENE | BOOLEAN | 60%

As the prospective twin, on the TCGA-105 overlap cohort. AUTHORED

values: `False (633) Â· True (299)`

example: `CDH1 â†’ True`

---

cptac_t105_agg_ret_prot
GENE | REAL, SIGNED | 42%

As the prospective twin, TCGA-105. AUTHORED

example: `CDH1 â†’ 0.2276`

---

cptac_t105_agg_rho_disc
GENE | SIGNED Âˆ’1Â€¦1 | AGG ARM | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against the discovery layer DERIVED

example: `CDH1 â†’ 0.04722`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_agg_rho_disc_raw
GENE | SIGNED Âˆ’1Â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against the discovery layer Â· UNADJUSTED â€” no confounder block DERIVED

example: `CDH1 â†’ -0.03473`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_agg_rho_prot
GENE | SIGNED Âˆ’1Â€¦1 | AGG ARM | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against protein DERIVED

example: `CDH1 â†’ -0.02019`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_agg_rho_prot_raw
GENE | SIGNED Âˆ’1Â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against protein Â· UNADJUSTED â€” no confounder block DERIVED

example: `CDH1 â†’ -0.08874`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_agg_rho_rna
GENE | SIGNED Âˆ’1Â€¦1 | AGG ARM | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against mRNA DERIVED

example: `CDH1 â†’ -0.07787`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_agg_rho_rna_raw
GENE | SIGNED Âˆ’1Â€¦1 | 60%

CPTAC TCGA-overlap cohort (n=105) Â· Î²-weighted aggregate Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€” no confounder block DERIVED

example: `CDH1 â†’ -0.1085`

> *Defined on: genes/edges present in the CPTAC cohort*

---

cptac_t105_n_arms
GENE | COUNT | 85%

Arms behind the TCGA-105 aggregate (median 2, max 57). See the prospective twin. AUTHORED

example: `CDH1 â†’ 21`

> *Defined on: genes/edges present in the CPTAC cohort*

DOMINANT_ Â· 4

reference â€" context, provenance and rarely-quoted detail

The top-ranked regulator of this gene in one state (HLY / NAT / TUM), by GLOBAL dose rank. The inputs to `regulatory_handoff`.

4 columns Â· rung **gene** Â· coverage **77%â€"83%** (median 82%) Â· **3** share the block description

3× identifier Â· 1× categorical

dominant_HLY
GENE IDENTIFIER 77%

in the GTEx-healthy state DERIVED

example: CDH1 â†' hsa-miR-92a-3p

dominant_NAT
GENE IDENTIFIER 81%

in matched normal (NAT) DERIVED

example: CDH1 â†' hsa-miR-30a-5p

dominant_TUM
GENE IDENTIFIER 82%

The arm carrying the largest share of the gene's tumour pressure budget (miR-21-5p on 56 genes, miR-200c-3p next). AUTHORED

example: CDH1 â†' hsa-miR-30a-5p

dominant_edge_shift_class
GENE CATEGORICAL 83%

per edge Â· shift Â· class label DERIVED

example: CDH1 â†' uncoupled

STATIC_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **87%â€"87%** (median 87%)

1× identifier

static_owner_family
GENE IDENTIFIER 87%

The static (dose-based) owner of the gene, for comparison against the realization owner. AUTHORED

example: CDH1 â†' miR-25-3p/32-5p/92-3p/363-3p/367-3p

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

TOTAL_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)

1× real â‰¥ 0

total_pressure
GENE REAL â‰¥ 0 100%

Sum over the gene's regulators. AUTHORED

example: CDH1 â†' 1.353

OWNER_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **41%â€"41%** (median 41%)

1× boolean

owner_agrees
GENE BOOLEAN 41%

Whether the realization owner and the static owner agree. Defined only where both exist. AUTHORED

values: False (385) Â· True (253)

example: CDH1 â†' True

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

MEDIAN_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€“100%** (median 100%)
1× real â‰¥ 0

median_retention
GENE | REAL â‰¥ 0 | 100%

A per-row median of a lower-rung quantity; check `agg_of`. AUTHORED
example: CDH1 â†' 0.9148

ARMRES_ Â· 7

reference â€” context, provenance and rarely-quoted detail

â IS THIS GENE MODELABLE AT ARM RESOLUTION â€” the gene-level roll-up of arm identifiability (`card_ladders.gene_arm_resolution`, 2026-08-19).
7 columns Â· rung **gene** Â· coverage **18%â€“18%** (median 18%) Â· **4** share the block description
2× signed â^'1â€¦1 Â· 2× count Â· 1× categorical Â· 1× fraction 0â€“1 Â· 1× real â‰¥ 0

armres_best_drho
GENE | SIGNED Â^'1Â€¦1 | 18%

the best DERIVED
example: CDH1 â†' -0.004735

armres_class
GENE | CATEGORICAL | 18%

`arm_modelable` (every multi-arm cell resolvable, 68 genes) Â· `partial` (37) Â· `family_only` (no cell resolvable, 168). Absent = the gene has no multi-arm cell at all, so the question is undefined. AUTHORED
values: family_only (168) Â· arm_modelable (68) Â· partial (37)
example: CDH1 â†' arm_modelable

armres_frac_resolvable
GENE | FRACTION 0Â€“1 | 18%

Fraction of this gene's multi-arm cells where the arms are separable. AUTHORED
example: CDH1 â†' 1

armres_med_drho
GENE | SIGNED Â^'1Â€¦1 | 18%

Median `oof_drho` over the gene's multi-arm cells â€” negative means arm resolution predicts repression better out of fold. AUTHORED
example: CDH1 â†' -0.004735

armres_med_sep_z
GENE | REAL Â‰¥ 0 | 18%

median Â· separation Â· z-score DERIVED
example: CDH1 â†' 3.196

armres_n_arms_split
GENE | COUNT | 18%

count Â· arms DERIVED
example: CDH1 â†' 2

armres_n_multi_cells
GENE | COUNT | 18%

count DERIVED
example: CDH1 â†' 1

IDENTITY_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **87%â€“87%** (median 87%)
1× boolean

identity_eq_magnitude
GENE | BOOLEAN | AGG FAMILY | 87%

Do the identity and magnitude answers name the SAME top family. Exactly 100% agreement at n_fam==1 by construction; 76% at n_fam>=2. Read it only on multi-family genes. AUTHORED
values: True (1,161) Â· False (191)
example: CDH1 â†' True

HEUR_ Â· 5

reference â€” context, provenance and rarely-quoted detail

5 columns Â· rung **gene** Â· coverage **89%â€“91%** (median 91%)
2× signed âˆ’1â€¦1 Â· 1× count Â· 1× boolean Â· 1× categorical

heur_delta_tumor_nat
GENE | SIGNED Âˆ'1Â€¦1 | 89%

Tumour-minus-NAT change in the heuristic pressure correlation. AUTHORED

example: CDH1 â†' 0.03725

heur_n_regulators
GENE | COUNT | 91%

âš âš **A DIFFERENT PIPELINE'S COUNT â€" not the model's.** It is `edges.groupby('gene')['miRNA'].nunique()` inside `analyses/misc/mirna_comovement.py`, i.e. AUTHORED

example: CDH1 â†' 12

heur_net_repressed_tumor
GENE | BOOLEAN | 91%

Is the gene net-repressed in tumour, per the RETIRED heuristic lane. True for 282 of 1,409 genes. AUTHORED

values: False (1,127) Â· True (282)

example: CDH1 â†' False

heur_repression_class
GENE | CATEGORICAL | 91%

Cross-state repression class from the RETIRED heuristic lane (never_repressed / lost_repression / gained_repression / constitutive_repressed). 867 of 1,409 genes are `never_repressed`. AUTHORED

values: never_repressed (867) Â· lost_repression (258) Â· gained_repression (167) Â· constitutive_repressed (115) Â· na (2)

example: CDH1 â†' never_repressed

heur_rho_pressure_tumor
GENE | SIGNED Âˆ'1Â€¦1 | 91%

Spearman of the gene's HEURISTIC aggregate pressure against its mRNA in tumour. AUTHORED

example: CDH1 â†' 0.05168

GREAL_ Â· 10

reference â€” context, provenance and rarely-quoted detail

Realization ladder for the GENE â€" the mirror of `real_`: how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). âœ… Cross-rung control: the gene-side totals reproduce the arm-side exactly.

10 columns Â· rung **gene** Â· coverage **29%â€“83%** (median 83%) Â· **8** share the block description
7× count Â· 2× signed âˆ’1â€¦1 Â· 1× fraction 0â€“1

greal_best_coupling
GENE | SIGNED Âˆ'1Â€¦1 | 83%

the best DERIVED

example: CDH1 â†' -0.271

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_frac_c10
GENE | FRACTION 0Â€"1 | 29%

Gene-rung twin of `real_frac_c10` â€" defined on 446 genes against `greal_n_c10`'s 1,289; gated-out genes have a median `greal_n_scored` of **1**. Exact on 100%. AUTHORED

example: CDH1 â†' 0.4

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_med_coupling
GENE | SIGNED Âˆ'1Â€¦1 | 83%

median DERIVED

example: CDH1 â†' -0.083

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

greal_n_c05
GENE COUNT 83%

count Â· at the â ˆ’0.05 threshold DERIVED

example: `CDH1 â†’ 7`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

---

greal_n_c10
GENE COUNT 83%

count Â· at the â ˆ’0.10 threshold DERIVED

example: `CDH1 â†’ 4`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

---

greal_n_c15
GENE COUNT 83%

count Â· at the â ˆ’0.15 threshold DERIVED

example: `CDH1 â†’ 2`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

---

greal_n_c20
GENE COUNT 83%

count Â· at the â ˆ’0.20 threshold DERIVED

example: `CDH1 â†’ 2`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

---

greal_n_c30
GENE COUNT 83%

count Â· at the â ˆ’0.30 threshold DERIVED

example: `CDH1 â†’ 0`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

---

greal_n_scored
GENE COUNT 83%

Genes/edges scored for realized coupling (median 2, max 59) â€" the DENOMINATOR of every `greal_frac_*` . âœ… The `greal_n_c05 â€¦ n_c30` ladder is properly nested: **0 monotonicity violations and 0 rows where a threshold count exceeds `n_scored`** (verified). AUTHORED

example: `CDH1 â†’ 10`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

---

greal_n_sig
GENE COUNT 83%

count DERIVED

example: `CDH1 â†’ 2`

> *Defined on: genes with â‰¥1 calibrated-coupling-scored regulator (1,260) â€" the GENE-side ladder; rates gated nâ‰¥3*

ACQUIRED_VS_ Â· 2

reference â€" context, provenance and rarely-quoted detail

> Acquisition of regulatory dose relative to a healthy reference (GTEx or NAT).
>
> **2** columns Â· rung **gene** Â· coverage **91%â€"91%** (median 91%) Â· **2** share the block description
>
> 2× real, signed

acquired_vs_gtex
GENE REAL, SIGNED 91%

GTEx DERIVED

example: `CDH1 â†’ 0.04213`

---

acquired_vs_nat
GENE REAL, SIGNED 91%

in matched normal (NAT) DERIVED

example: `CDH1 â†’ 0.3754`

PRESSURE_ Â· 1

reference â€" context, provenance and rarely-quoted detail

> **1** columns Â· rung **gene** Â· coverage **91%â€"91%** (median 91%)
>
> 1× real â‰¥ 0

pressure_tumor

GENE | REAL Â‰¥ 0 | 91%

Aggregate incoming pressure on the gene in the named state. AUTHORED

example: CDH1 â†’ 3.784

REALIZATION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **42%â€“42%** (median 42%)
1× identifier

realization_owner

GENE | IDENTIFIER | 42%

Within-patient realization quantities â€” the score, its identity allocation, and who owns it. AUTHORED

example: CDH1 â†’ miR-25-3p/32-5p/92-3p/363-3p/367-3p

> *Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)*

TCGA_ Â· 4

reference â€” context, provenance and rarely-quoted detail

The TCGA mRNA leg carried beside the CPTAC columns so the two are comparable on the same gene rows.

4 columns Â· rung **gene** Â· coverage **89%â€“89%** (median 89%) Â· **3** share the block description
3× signed â ’1â€¦1 Â· 1× count

tcga_abund_rho_rna

GENE | SIGNED Â ’1Â€¦1 | AGG ARM | 89%

unweighted abundance-sum reference Â· Spearman Ï Â· against mRNA DERIVED

example: CDH1 â†’ 0.006381

tcga_agg_rho_rna

GENE | SIGNED Â ’1Â€¦1 | AGG ARM | 89%

Gene-level Î²-weighted aggregate coupling to mRNA in TCGA (median **â ’0.05**) against `tcga_abund_rho_rna`'s **â ’0.02** â€” the abundance-sum reference. AUTHORED

example: CDH1 â†’ -0.1922

tcga_agg_rho_rna_raw

GENE | SIGNED Â ’1Â€¦1 | 89%

Î²-weighted aggregate Â· Spearman Ï Â· against mRNA Â· UNADJUSTED â€” no confounder block DERIVED

example: CDH1 â†’ -0.1921

tcga_n_arms

GENE | COUNT | 89%

count Â· arms DERIVED

example: CDH1 â†’ 23

DISC_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **gene** Â· coverage **3%â€“100%** (median 51%)
1× categorical Â· 1× count

disc_gold_families

GENE | CATEGORICAL | 3%

Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

example: KLF3 â†’ miR-130-3p/301-3p/454-3p;miR-17-5p/20-5p/93-5p/106-5p/519-3p

disc_n_gold_edges

GENE | COUNT | 100%

How many of the 157 discovery-queue edges name this gene (max 4). AUTHORED

values: 0 (1,510) Â· 1 (26) Â· 3 (5) Â· 2 (5) Â· 4 (3)

example: CDH1 → 0

MAX_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€“100%** (median 100%)
1× fraction 0â€“1

max_beta_frac_sd
GENE FRACTION 0Â€“1 100%

Maximum over the row's constituent units. AUTHORED
example: CDH1 â†' 0.03738

REGULATORY_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€“100%** (median 100%)
1× boolean

regulatory_handoff
GENE BOOLEAN 100%

Does the gene's globally-dominant regulator differ between healthy and tumour OR between NAT and tumour. AUTHORED
values: False (1,221) Â· True (328)
example: CDH1 â†' True

MEAN_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **82%â€“82%** (median 82%)
1× real, signed

mean_dTarget
GENE REAL, SIGNED 82%

A per-row mean of a lower-rung quantity; check `agg_of` for which rung it summarises. AUTHORED
example: CDH1 â†' 1.18

CONCENTRATION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **53%â€“53%** (median 53%)
1× fraction 0â€“1

concentration_adj
GENE FRACTION 0Â€“1 53%

`(concentration âˆ' 1/n_fam) / (1 âˆ' 1/n_fam)` âˆˆ [0,1] â€” how concentrated the gene is RELATIVE to the most diffuse its own design allows. AUTHORED
example: CDH1 â†' 0.1573

FAM_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **81%â€“81%** (median 81%)
1× signed âˆ'1â€¦1

fam_pooled_rho_adj
GENE SIGNED Âˆ'1Â€¦1 81%

Seed-family structure the arm sits in â€” membership, affinity and dominance within it. AUTHORED
example: CDH1 â†' -0.096

gene_family_card.tsv

gene_family

One row per one (seed family, gene) pair. Keyed [gene, family] â€" it cannot express a property of the family itself. That is the seed_family card's job.

61 columns **5,117** rows **12** blocks median coverage **97%**

Contents â€" 12 blocks, most important first

(BARE) Â· 8

core â€" the columns findings rest on

8 columns Â· **spans family, key** â€" this prefix groups columns living on DIFFERENT units Â· coverage **95%â€"100%** (median 100%)
3× identifier Â· 3× real â‰¥ 0 Â· 1× boolean Â· 1× real, signed

arms
FAMILY IDENTIFIER 100%

The arms collapsed into this (gene, family) cell, **semicolon-separated** (`hsa-miR-17-5p;hsa-miR-20a-5p`). AUTHORED
example: `CDH1 / miR-132-3p/212-3p â†' hsa-miR-132-3p`

beta
FAMILY REAL â‰¥ 0 100%

FAMILY-rung coupling coefficient â€" `readouts.run(level='family')`, the same-seed collapse APPLIED. AUTHORED
example: `CDH1 / miR-132-3p/212-3p â†' 0.07163`

family
KEY IDENTIFIER 100%

KEY â€" seed family of the regulator, as used by the family-rung fit. AUTHORED
example: `CDH1 â†' miR-132-3p/212-3p`

gene
KEY IDENTIFIER 100%

KEY â€" HGNC gene symbol. AUTHORED
example: `miR-9-3p â†' ITGB1`

identified
FAMILY BOOLEAN 100%

`|z| > 2` on the UNCALIBRATED SD â€" **27.1%** of cells. `cal_identified` is the honest version. AUTHORED
values: False (3,730) Â· True (1,387)
example: `CDH1 / miR-132-3p/212-3p â†' False`

identity
FAMILY REAL, SIGNED 95%

Shapley/LMG identity at the FAMILY rung â€" same estimator as the edge card's `identity`, different unit. AUTHORED
example: `CDH1 / miR-132-3p/212-3p â†' 0.0641`

retention
FAMILY REAL â‰¥ 0 100%

`beta_deconv / beta_core` at the family grain â€" the ADJUSTMENT sense of retention (what fraction of the coupling survives the composition block), never the patient-baseline sense. AUTHORED
example: `CDH1 / miR-132-3p/212-3p â†' 0.8516`

z
FAMILY | REAL Â‰¥ 0 | 100%

`beta / beta_sd` for the family cell, on the **UNCALIBRATED** posterior SD. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 1.844`

N_ Â· 3

supporting â€" read these when the core raises a question

> **3** columns Â· **spans family, gene** â€" this prefix groups columns living on DIFFERENT units Â· coverage **100%â€"100%** (median 100%)
> 2× count Â· 1× constant

n_arms
FAMILY | COUNT | 100%

How many arms of THIS family sit in THIS gene's design. âœ... FAMILY rung, VERIFIED (2026-08-19): it varies within gene for 241 of 1,549 genes, and the per-gene SUM of these equals the gene card's `n_arms` exactly â€" a clean cross-rung consistency check. AUTHORED

values: 1 (4,646) Â· 2 (338) Â· 3 (87) Â· 4 (25) Â· 5 (11) Â· 6 (6) Â· 7 (4)

example: `CDH1 / miR-132-3p/212-3p â†' 1`

n_fam
GENE | COUNT | 100%

How many families compete at this gene. beta is mathematically INERT at 1. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 25`

n_samples
GENE | CONSTANT | 100%

Samples the family cell was fit on â€" **CONSTANT 1,040** across the card. AUTHORED

values: 1040 (5,117)

example: `CDH1 / miR-132-3p/212-3p â†' 1040`

CTX_ Â· 16

supporting â€" read these when the core raises a question

> Gene-design context from the decoy/atlas layer (`learned/analyses/gene_atlas.py` via `card_context`): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
>
> **16** columns Â· rung **gene** Â· SUBSET of the edge block, identical rungs Â· coverage **81%â€"100%** (median 96%) Â· **13** share the block description
> 4× signed âˆ'1â€¦1 Â· 3× real â‰¥ 0 Â· 3× real, signed Â· 3× count Â· 1× categorical Â· 1× fraction 0â€"1 Â· 1× boolean

ctx_apriori_class
GENE | CATEGORICAL | 100%

class label DERIVED

values: A_COMPETENT (3,402) Â· D_UNDEFINED (730) Â· B_PARTIAL (590) Â· C_WEAK (395)

example: `CDH1 / miR-132-3p/212-3p â†' A_COMPETENT`

ctx_ceiling
GENE | SIGNED Âˆ'1Â€¦1 | 99%

The UPPER BOUND on what ANY estimator can achieve for this gene â€" the 5-fold out-of-fold RÂ² of an UNRESTRICTED OLS on the gene's real design (`card_context.py:15`). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.1535`

ctx_collapse
GENE | REAL Â‰¥ 0 | 96%

`n_arm / n_fam` for the gene â€" **how much curation BUNDLES same-seed mates onto it** (1â€"5). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 1.091`

ctx_d_collin
GENE | SIGNED Âˆ'1Â€¦1 | 81%

change Â· collinearity DERIVED

example: `CDH1 / miR-132-3p/212-3p â†' -0.02497`

> *Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)*

ctx_d_fam
GENE | REAL, SIGNED | 95%

change Â· per seed family DERIVED

example: `CDH1 / miR-132-3p/212-3p â†' 0`

ctx_dose_max
GENE · REAL Â‰¥ 0 · 96%

dose Â· maximum DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 15.38

ctx_dose_med
GENE · REAL Â‰¥ 0 · 96%

dose Â· median DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 6.318

ctx_frac_abund
GENE · FRACTION 0Â€“1 · 96%

fraction Â· unweighted abundance-sum reference DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 0.5833

ctx_gap_core
GENE · SIGNED Âˆ’1Â€¦1 · 95%

gap Â· core confounder block DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ -0.2714

ctx_gap_d_dose
GENE · REAL, SIGNED · 95%

gap Â· change Â· dose DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 0.1238

ctx_gap_deconv
GENE · SIGNED Âˆ’1Â€¦1 · 95%

gap Â· composition-adjusted (deconvolution block) DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ -0.1864

ctx_gap_retention
GENE · REAL, SIGNED · 92%

gap Â· retention DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 0.6867

ctx_measurable
GENE · BOOLEAN · 99%

Is the gene's `ctx_ceiling` above `CEILING_ROBUST` (**0.02**, not 0 â€” MH-144)? AUTHORED
values: True (3,652) Â· False (1,432)
example: CDH1 / miR-132-3p/212-3p â†’ True

ctx_n_abund
GENE · COUNT · 96%

count Â· unweighted abundance-sum reference DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 7

ctx_n_fam_abund
GENE · COUNT · 96%

count Â· per seed family Â· unweighted abundance-sum reference DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 6

ctx_n_fam_multi
GENE · COUNT · 96%

count Â· per seed family DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 1

BETA_ Â· 7

supporting â€” read these when the core raises a question

> Posterior summaries of the coupling coefficient from the dense Gibbs fit (`learned/readouts.py`). `_sd` is the posterior SD, `_frac` the share of the gene's total beta.
>
> **7** columns Â· rung **family** Â· coverage **100%â€“100%** (median 100%) Â· **7** share the block description
> 4× fraction 0â€“1 Â· 3× real â‰¥ 0

beta_deconv
FAMILY · REAL Â‰¥ 0 · 100%

composition-adjusted (deconvolution block) DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 0.061

beta_frac
FAMILY · FRACTION 0Â€“1 · 100%

fraction DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 0.05285

beta_frac_abs
FAMILY · FRACTION 0Â€“1 · 100%

fraction DERIVED
example: CDH1 / miR-132-3p/212-3p â†’ 0.05296

beta_frac_deconv
FAMILY | FRACTION 0â€“1 | 100%

fraction Â· composition-adjusted (deconvolution block) DERIVED
example: CDH1 / miR-132-3p/212-3p â†' 0.05107

beta_frac_sd
FAMILY | FRACTION 0â€“1 | 100%

fraction Â· standard deviation DERIVED
example: CDH1 / miR-132-3p/212-3p â†' 0.02825

beta_sd
FAMILY | REAL â‰¥ 0 | 100%

standard deviation DERIVED
example: CDH1 / miR-132-3p/212-3p â†' 0.03885

beta_sd_deconv
FAMILY | REAL â‰¥ 0 | 100%

standard deviation Â· composition-adjusted (deconvolution block) DERIVED
example: CDH1 / miR-132-3p/212-3p â†' 0.03579

COMP_ Â· 16

reference â€” context, provenance and rarely-quoted detail

> Cell-composition drivers of this gene's coupling (`compartment_*`): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
>
> **16** columns Â· rung **gene** Â· coverage **66%â€“97%** (median 81%) Â· **2** share the block description
> 6× categorical Â· 6× signed â^'1â€¦1 Â· 3× real, signed Â· 1× real â‰¥ 0

comp_cptac_prot_driver
GENE | CATEGORICAL | 66%

WHICH deconvolved compartment most explains this gene's PROTEIN coupling â€" `prolif` 219 Â· `CAFs` 173 Â· `purity` 140 Â· `PVL` 77 Â· `Endothelial` 67. AUTHORED
example: CDH1 / miR-132-3p/212-3p â†' T-cells

comp_cptac_prot_driver_ret
GENE | REAL, SIGNED | 66%

`1 â^' comp_cptac_prot_driver_share`, exactly. Median +0.398 raw, **+0.469** gated to |Ï_raw| â‰¥ 0.10 â€" quote the gated value. See the mRNA twin. AUTHORED
example: CDH1 / miR-132-3p/212-3p â†' 0.5085

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_driver_share
GENE | REAL, SIGNED | 66%

Protein-layer twin of `comp_tcga_mrna_driver_share`. Same exact complementarity with `_driver_ret` (8.9e-16 on 918 rows). AUTHORED
example: CDH1 / miR-132-3p/212-3p â†' 0.4915

comp_cptac_prot_rho_raw
GENE | SIGNED â^'1â€¦1 | 81%

Unadjusted aggregate PROTEIN coupling â€" denominator of `comp_cptac_prot_driver_ret`. AUTHORED
example: CDH1 / miR-132-3p/212-3p â†' -0.1416

comp_cptac_prot_top_budget
GENE | CATEGORICAL | 81%

CPTAC Â· against protein Â· the top-ranked Â· pressure budget DERIVED
example: CDH1 / miR-132-3p/212-3p â†' Normal Epithelial

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_budget_r
GENE | SIGNED â^'1â€¦1 | 81%

As the TCGA mRNA twin, on CPTAC protein â€" the budget-vs-compartment correlation. Median **â^'0.118**, i.e. it runs the OPPOSITE way to mRNA (+0.121). AUTHORED
example: CDH1 / miR-132-3p/212-3p â†' 0.3126

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_target
GENE | CATEGORICAL | 81%

CPTAC Â· against protein Â· the top-ranked Â· the target gene's expression DERIVED
example: CDH1 / miR-132-3p/212-3p â†' T-cells

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

comp_cptac_prot_top_target_r
GENE | SIGNED Âˆ'1Â€¦1 | 81%

As the TCGA mRNA twin, on CPTAC protein â€" the target-vs-compartment correlation. Median **+0.203**. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' -0.2385`

> *Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)*

---

comp_tcga_mrna_driver
GENE | CATEGORICAL | 81%

WHICH compartment drives the gene's mRNA coupling â€" the one whose removal moves it most (`CAFs` 279 Â· `purity` 174 Â· `prolif` 148 Â· `Endothelial` 122 Â· `Myeloid` 109). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' T-cells`

---

comp_tcga_mrna_driver_ret
GENE | REAL, SIGNED | 81%

`1 âˆ' comp_tcga_mrna_driver_share`, exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.8681`

---

comp_tcga_mrna_driver_share
GENE | REAL â‰¥ 0 | 81%

Share of the gene's mRNA coupling attributable to the single compartment that most changes it â€" **axiom 8's composition GATE**, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.1319`

---

comp_tcga_mrna_rho_raw
GENE | SIGNED Âˆ'1Â€¦1 | 97%

The gene's UNADJUSTED aggregate mRNA coupling â€" the DENOMINATOR of `comp_tcga_mrna_driver_ret`. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' -0.1921`

---

comp_tcga_mrna_top_budget
GENE | CATEGORICAL | 97%

The compartment the gene's miRNA BUDGET loads on most â€" a property of the REGULATORS. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' Normal Epithelial`

---

comp_tcga_mrna_top_budget_r
GENE | SIGNED Âˆ'1Â€¦1 | 97%

Correlation between the gene's **miRNA PRESSURE BUDGET** and the fraction of the compartment that budget loads on most (`comp_tcga_mrna_top_budget`), in TCGA mRNA. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.2545`

---

comp_tcga_mrna_top_target
GENE | CATEGORICAL | 97%

The compartment the TARGET GENE's expression loads on most. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' prolif`

---

comp_tcga_mrna_top_target_r
GENE | SIGNED Âˆ'1Â€¦1 | 97%

Correlation between the **TARGET GENE's own expression** and the fraction of the compartment it loads on most. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.3289`

IDENTITY_ Â· 4

reference â€" context, provenance and rarely-quoted detail

> **4** columns Â· rung **family** Â· coverage **95%â€"97%** (median 95%)
> 2× fraction 0â€"1 Â· 1× real, signed Â· 1× boolean

identity_abs
FAMILY | FRACTION 0Â€"1 | 95%

|identity| / Î£|identity| per gene â€" the BOUNDED companion to `identity`, always in [0,1] (verified: 0 non-NaN values outside it on either card). AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.063`

---

identity_coherence
FAMILY | FRACTION 0Â€"1 | 95%

1 / Î£|identity| per GENE, in (0,1] â€" identity's own sign coherence. 1.0 when nothing cancels, â†' 0 as suppressor cancellation grows. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.983`

identity_deconv

FAMILY | REAL, SIGNED | 97%

Shapley/LMG attribution on R^2 with bagged-NNLS weights â€” WHO regulates the gene, collinearity-fair. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.04796`

---

identity_reliable

FAMILY | BOOLEAN | 95%

`identity_coherence â‰¥ 0.5` AND `|identity| â‰¤ 1`. Admits **93.3%** of edge rows and **92.6%** of gene_family rows, and excludes **every** row whose |identity| exceeds 1 (119/119 and 118/118). AUTHORED

values: `True (4,740) Â· False (146)`

example: `CDH1 / miR-132-3p/212-3p â†' True`

PIP_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× constant Â· 1× fraction 0â€"1

pip_dense

FAMILY | CONSTANT | 100%

âš CONSTANT by construction (Ï€â‰¡1 dense readout). AUTHORED

values: `1.0 (5,117)`

example: `CDH1 / miR-132-3p/212-3p â†' 1`

---

pip_discovery

FAMILY | FRACTION 0â€"1 | 100%

Posterior inclusion probability. `pip_dense` comes from the pi=1 coupling readout; `pip_discovery` from the evidence-pi readout. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.3261`

W_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **gene** Â· coverage **100%â€"100%** (median 100%)
1× real â‰¥ 0

w_max

GENE | REAL â‰¥ 0 | 100%

Maximum curated EVIDENCE weight for this (gene, family) cell. Family-rung here; gene-rung on the gene card. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 13.24`

M_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× real â‰¥ 0

m_nnls

FAMILY | REAL â‰¥ 0 | 100%

Bagged-NNLS weight for this (gene, family) cell. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†' 0.07211`

COMPOSITION_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **100%â€"100%** (median 100%)
1× categorical

composition_class

FAMILY CATEGORICAL 100%

Composition-adjustment verdict for this row (e.g. `composition_explained` when the effect does not survive the deconvolution block). AUTHORED

values: `cell_intrinsic (3,378) Â· partial (1,095) Â· composition_explained (644)`

example: `CDH1 / miR-132-3p/212-3p â†’ cell_intrinsic`

PRIOR_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **100%â€“100%** (median 100%)
1× fraction 0â€“1

prior_pi

FAMILY FRACTION 0â€“1 100%

The evidence prior fed to the discovery readout. AUTHORED

example: `CDH1 / miR-132-3p/212-3p â†’ 0.2962`

RETENTION_ Â· 1

reference â€” context, provenance and rarely-quoted detail

1 columns Â· rung **family** Â· coverage **100%â€“100%** (median 100%)
1× boolean

retention_reliable

FAMILY BOOLEAN 100%

Fraction of an effect surviving covariate adjustment (adjusted / raw). AUTHORED

values: `False (3,730) Â· True (1,387)`

example: `CDH1 / miR-132-3p/212-3p â†’ False` arm_card.tsv

arm

One row per the miRNA arm itself, gene-free. The deepest card. An arm-rung column must be constant within arm ACROSS genes; if it varies by gene it is not an arm property, whatever its name says.

296 columns **2,450** rows **44** blocks median coverage **30%**

Contents â€” 44 blocks, most important first

ADM_ Â· 4

core â€” the columns findings rest on

4 columns Â· rung **arm** Â· AGGREGATE of the edge block â€" verified: n_with_site and n_admissible reproduce a direct rollup on 100% of arms (n_edges only 77.5%, the MH-252 universe gap) Â· coverage **14%â€"29%** (median 29%)
3× count Â· 1× fraction 0â€"1

adm_frac_with_site
ARM FRACTION 0â€"1 14%

`adm_n_with_site / adm_n_edges`. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9429

adm_n_admissible
ARM COUNT 29%

Edges of this arm that pass admissibility (`adm_has_site` + the evidence bar) â€" MH-216's single most load-bearing conditioning variable, rolled up per arm. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 33

adm_n_edges
ARM | COUNT | 29%

ADMISSIBILITY â€” could this edge repress in breast AT ALL? `has_site` (seed match in the 3'UTR) AND `expressed` (arm present in TCGA tumour OR GTEx healthy breast). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 35`

---

adm_n_with_site
ARM | COUNT | 29%

Edges of this arm whose target carries a seed site â€" **89.9% of edges (4,436/4,937)**, so the site requirement is NOT the binding constraint; the evidence bar is (admissible drops it to 60.4%). 19.7% of arms have none. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 33`

> *Defined on: edges/arms with an admissibility tag (4,937 edges; has_site 89.9% Â· expressed 63.9% Â· admissible 60.4%)*

ARM_ Â· 7

core â€" the columns findings rest on

> **7** columns Â· rung **arm** Â· coverage **15%â€"20%** (median 20%)
> 4× real, signed Â· 2× real â‰¥ 0 Â· 1× percent

arm_iqr
ARM | REAL â‰¥ 0 | 20%

Interquartile range of the arm's tumour RPM â€" its DYNAMIC RANGE. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1.1`

---

arm_lfc_HLY_NAT_raw
ARM | REAL, SIGNED | 15%

log-FC GTEx-healthy â†' TCGA-NAT, raw. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 7.94`

---

arm_lfc_HLY_TUM_QN
ARM | REAL, SIGNED | 16%

log-FC GTEx-healthy â†' TCGA-tumour, on quantile-normalised values. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 5.11`

---

arm_lfc_HLY_TUM_raw
ARM | REAL, SIGNED | 15%

As `arm_lfc_HLY_TUM_QN` but against the RAW GTEx median. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 10`

---

arm_lfc_NAT_TUM
ARM | REAL, SIGNED | 20%

log-FC of the arm's median abundance, matched NAT â†' tumour. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 2.06`

---

arm_med_rpm
ARM | REAL â‰¥ 0 | 20%

The arm's MEDIAN RPM across tumour samples â€" its abundance level, gene-free. Defined on the 19.8% of arms with tumour expression data. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 13.4`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

---

arm_pct_above_floor
ARM | PERCENT | 20%

â›" **THE NAME READS BACKWARDS.** Computed as `100Â·mean(x > FLOOR)` â€" the percent of samples ABOVE the detection floor, i.e. AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 100`

SEED_ Â· 1

core â€" the columns findings rest on

> **1** columns Â· rung **arm** Â· coverage **100%â€"100%** (median 100%)
> 1× identifier

seed_family
ARM | IDENTIFIER | 100%

KEY â€" the seed family itself, one row per family. AUTHORED

example: `hsa-let-7a-5p â†' let-7-5p/98-5p`

HEALTHY_ Â· 3

core â€” the columns findings rest on

3 columns Â· rung **arm** Â· identical column set to the edge block Â· coverage **16%â€"20%** (median 16%)
1× categorical Â· 1× real â‰¥ 0 Â· 1× boolean

healthy_leg
ARM | CATEGORICAL | 20%

Provenance of this arm's GTEx healthy leg â€" `measured`, `collapse_blind` or `true_absent`. AUTHORED

values: measured (230) Â· true_absent (168) Â· collapse_blind (100)

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' measured

healthy_potential
ARM | REAL Â‰¥ 0 | 16%

The arm's IMPUTED healthy abundance â€" a repression-POTENTIAL proxy, collapse-repaired (0.27 â€" 17.9). AUTHORED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 8.24

healthy_uninformative
ARM | BOOLEAN | 16%

Is the healthy leg both `collapse_blind` AND high-potential â€" i.e. the one combination where an *acquired* call is genuinely unsafe? **True on 143 of 3,490 edges (4%).** AUTHORED

values: False (392) Â· True (7)

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' False

SITE_ Â· 12

supporting â€" read these when the core raises a question

Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.

12 columns Â· rung **arm** Â· coverage **29%â€"29%** (median 29%) Â· **7** share the block description
7× count Â· 2× fraction 0â€"1 Â· 2× real, signed Â· 1× real â‰¥ 0

site_frac_8mer
ARM | FRACTION 0â€"1 | 29%

8mer share of the arm's **CANONICAL** sites (`site_n_8mer / site_n_canonical`; median 0.150). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.2169

site_frac_canonical
ARM | FRACTION 0â€"1 | 29%

Canonical share of the arm's **TOTAL** sites (`site_n_canonical / site_n_total`; median 0.054) â€" a different denominator from `site_frac_8mer`. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.03913

site_n_6mer
ARM | COUNT | AGG EDGE | 29%

count Â· 6mer sites DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1459

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

site_n_7mer
ARM | COUNT | AGG EDGE | 29%

count Â· 7mer sites DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 278

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

site_n_8mer
ARM | COUNT | AGG EDGE | 29%

count Â· 8mer sites DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 77

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

site_n_canonical
ARM | COUNT | AGG EDGE | 29%

count Â· canonical sites DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 355

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

site_n_genes_canonical
ARM | COUNT | AGG EDGE | 29%

count Â· genes Â· canonical sites DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 283

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

site_n_noncanon

ARM COUNT AGG EDGE 29%

count Â· non-canonical sites DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 7258

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

---

site_n_total

ARM COUNT AGG EDGE 29%

count Â· total DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 9072

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

---

site_repression_med

ARM REAL, SIGNED AGG EDGE 29%

Median predicted repression score over the arm's sites (all negative by construction; median âˆ'0.316). A PREDICTION from sequence, not a measurement â€" never cite it as evidence of observed repression. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' -0.3859

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

---

site_repression_min

ARM REAL, SIGNED AGG EDGE 29%

Strongest (most negative) predicted repression among the arm's sites (median âˆ'1.964). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' -2.437

---

site_sites_per_gene

ARM REAL â‰¥ 0 29%

Canonical sites per targeted gene (median 1.34) â€" site DENSITY, separating an arm with many sites on few genes from one spread thin. `site_n_canonical / site_n_genes_canonical`. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1.254

> *Defined on: arms in the scanMiR site-type table (771) â€" âš HALLMARK-SCOPED, 1,432 genes only*

ARB_ Â· 16

supporting â€" read these when the core raises a question

> Per-ARM rollup of its edge set â€" how many edges and genes it regulates and its mean |beta| over them.
>
> **16** columns Â· rung **arm** Â· coverage **28%â€"30%** (median 30%) Â· **10** share the block description
> 12× count Â· 2× real â‰¥ 0 Â· 1× fraction 0â€"1 Â· 1× signed âˆ'1â€¦1

arb_frac_identified

ARM FRACTION 0â€"1 AGG EDGE 30%

`arb_n_identified / arb_n_edges` (verified exact on 100% of rows). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1316

---

arb_max_abs_beta

ARM REAL â‰¥ 0 AGG EDGE 30%

Largest |Î²| among the arm's edges. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.2173

---

arb_max_identity

ARM SIGNED âˆ'1â€¦1 AGG EDGE 28%

Largest `identity` among this arm's edges, over reliable rows only. âœ… Now runs **âˆ'0.255 â€¦ +1.000** â€" it reached **+740.0** until the inert `beta_frac_reliable` gate was replaced (2026-08-19). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

---

arb_mean_abs_beta

ARM REAL â‰¥ 0 AGG EDGE 30%

Mean |Î²| over the arm's edges. See `arb_max_abs_beta` for the one-edge degeneracy (9.3% of arms, where the two are equal by construction). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.02837

> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_b001

ARM COUNT 30%

count Â· at the 0.001 threshold DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 18

> *Defined on: arms with a learned-Î² edge rollup (728)*

arb_n_b001_ident
ARM COUNT 30%

count Â· at the 0.001 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 5`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_b005
ARM COUNT 30%

count Â· at the 0.005 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_b005_ident
ARM COUNT 30%

count Â· at the 0.005 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 5`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_b010
ARM COUNT 30%

count Â· at the 0.010 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 4`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_b010_ident
ARM COUNT 30%

count Â· at the 0.010 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 4`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_b025
ARM COUNT 30%

count Â· at the 0.025 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_b025_ident
ARM COUNT 30%

count Â· at the 0.025 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_comp_explained
ARM COUNT AGG EDGE 30%

count DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_edges
ARM COUNT AGG EDGE 30%

Edges this arm carries in the LEARNED MODEL's own readouts. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 38`

---

arb_n_identified
ARM COUNT AGG EDGE 30%

count DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 5`
> *Defined on: arms with a learned-Î² edge rollup (728)*

---

arb_n_identity_reliable
ARM COUNT AGG EDGE 30%

Edges of this arm whose `identity` share is usable (`identity_reliable`). AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 33`

FAME_ Â· 28

supporting â€" read these when the core raises a question

> Study-attention axis for the arm â€" distinct PMIDs and distinct curated genes.
>
> **28** columns Â· rung **arm** Â· coverage **94%â€"96%** (median 96%) Â· **23** share the block description
> 27× count Â· 1× fraction 0â€"1

fame_assay_binding__functional_mti_weak_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· weak evidence Â· study count DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 150`
> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_binding_studies
ARM COUNT 96%

Studies whose assay class is BINDING (the marginal of the assay Ã— functional-class cross-tab). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 150`

---

fame_assay_functional_mti_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 35`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_functional_mti_weak_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· weak evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 158`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_nonfunctional_mti_studies
ARM COUNT 96%

by assay class Â· miRNAâ€"target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 13`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_other__functional_mti_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 4`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_other__functional_mti_weak_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· weak evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 2`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_other__nonfunctional_mti_studies
ARM COUNT 96%

by assay class Â· miRNAâ€"target interaction evidence Â· study count DERIVED

values: `0.0 (2,225) Â· 1.0 (77) Â· 2.0 (17) Â· 3.0 (10) Â· 5.0 (6) Â· 4.0 (4) Â· 6.0 (3) Â· 7.0 (1)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_other_studies
ARM COUNT 96%

by assay class Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_protein__functional_mti_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 28`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_protein__functional_mti_weak_studies
ARM COUNT 96%

by assay class Â· functional assays Â· miRNAâ€"target interaction evidence Â· weak evidence Â· study count DERIVED

values: `0.0 (2,225) Â· 1.0 (75) Â· 2.0 (22) Â· 3.0 (10) Â· 4.0 (6) Â· 5.0 (3) Â· 10.0 (1) Â· 25.0 (1)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_protein__nonfunctional_mti_studies
ARM COUNT 96%

by assay class Â· miRNAâ€"target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 10`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_protein_studies
ARM COUNT 96%

by assay class Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 31`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

fame_assay_reporter__functional_mti_studies

ARM | COUNT | 96%

by assay class Â· functional assays Â· miRNAâ€“target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 31`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_reporter__nonfunctional_mti_studies

ARM | COUNT | 96%

by assay class Â· miRNAâ€“target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 10`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_reporter_studies

ARM | COUNT | 96%

by assay class Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 34`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_rna__functional_mti_studies

ARM | COUNT | 96%

by assay class Â· against mRNA Â· functional assays Â· miRNAâ€“target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 22`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_rna__functional_mti_weak_studies

ARM | COUNT | 96%

by assay class Â· against mRNA Â· functional assays Â· miRNAâ€“target interaction evidence Â· weak evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 9`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_rna__nonfunctional_mti_studies

ARM | COUNT | 96%

by assay class Â· against mRNA Â· miRNAâ€“target interaction evidence Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 5`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_rna_studies

ARM | COUNT | 96%

by assay class Â· against mRNA Â· study count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 30`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_assay_total_studies

ARM | COUNT | 96%

Distinct STUDIES (`study_id.nunique()`) behind this arm's curated edges â€" median 57, max 930. See `fame_assay_unique_targets`: different definitions, one axis (spearman 1.0000). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 189`

---

fame_assay_unique_targets

ARM | COUNT | 96%

Distinct TARGET GENES this arm has in the curated literature (`gene.nunique()`). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 189`

---

fame_led_frac_ltfunc

ARM | FRACTION 0â€“1 | AGG EDGE | 96%

fraction DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.4565`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_led_n_classes

ARM | COUNT | AGG EDGE | 96%

How many distinct evidence CLASSES this arm's curated edges span (1â€“7). AUTHORED

values: `1.0 (1,466) Â· 5.0 (246) Â· 6.0 (241) Â· 4.0 (166) Â· 2.0 (137) Â· 3.0 (76) Â· 7.0 (18)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 6`

---

fame_led_n_ltfunc_pmid

ARM | COUNT | AGG EDGE | 96%

count Â· distinct publications DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 42`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_led_n_weak

ARM | COUNT | AGG EDGE | 94%

count Â· weak evidence DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 206`

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

fame_n_genes_curated
ARM · COUNT · AGG EDGE · 96%

count Â· genes DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 3216

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

---

fame_npmid
ARM · COUNT · AGG EDGE · 96%

Distinct PMIDs citing this arm. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 92

AGO_ Â· 4

supporting â€" read these when the core raises a question

---

4 columns Â· rung **arm** Â· âš arm.ago_dom and edge.ago_loading are RANK-IDENTICAL (spearman +1.000, equal on 73.2%) â€" one quantity, two names. Never scan both. Â· coverage **28%â€"60%** (median 28%)
1× fraction 0â€"1 Â· 1× constant Â· 1× categorical Â· 1× real â‰¥ 0

ago_dom
ARM · FRACTION 0â€"1 · 28%

Fraction of the arm's Manakov AGO-CLIP reads assigned to it rather than its sibling arm â€" the loading dominance. Median 0.658 on the 27.7% of arms with CLIP coverage. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9937

---

ago_dom_src
ARM · CONSTANT · 28%

Provenance of `ago_dom` â€" which CLIP dataset the AGO-loading dominance came from. AUTHORED
values: manakov (678)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' manakov

---

ago_guide_class
ARM · CATEGORICAL · 28%

AGO/RISC loading gate for the arm. AUTHORED
values: guide (288) Â· co_loaded (204) Â· passenger (186)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' guide

---

ago_reads
ARM · REAL â‰¥ 0 · 60%

Raw AGO-CLIP read count (median 6, max 210,722). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 9900

(BARE) Â· 1

supporting â€" read these when the core raises a question

---

1 columns Â· rung **key** Â· coverage **100%â€"100%** (median 100%)
1× identifier

arm
KEY · IDENTIFIER · 100%

KEY â€" mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through `_arm_key()`, which gates against the canonical universe and LOGS drops rather than silently dropping. AUTHORED
example: miR-588 â†' hsa-miR-4701-5p

HLY_ Â· 7

supporting â€" read these when the core raises a question

HEALTHY baseline levels for the arm â€" GTEx breast median/percentile and TCGA-NAT median.
7 columns Â· rung **arm** Â· coverage **17%â€"91%** (median 91%) Â· **3** share the block description
2× real â‰¥ 0 Â· 1× categorical Â· 1× boolean Â· 1× percent Â· 1× count Â· 1× fraction 0â€"1

hly_baseline_src
ARM | CATEGORICAL | 91%

source DERIVED

values: floor0 (1,137) Â· mited_anchor (634) Â· gtex (417) Â· gtex_family (49)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' mited_anchor

> *Defined on: arms with a GTEx healthy anchor â€" âš MEASURED leg only; floor0 is NEVER emitted as a value*

---

hly_from_seedmate
ARM | BOOLEAN | 91%

Was this arm's healthy baseline borrowed from a SEED-MATE rather than measured directly (49 arms)? AUTHORED

values: False (2,188) Â· True (49)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' False

---

hly_gtex_median
ARM | REAL Â‰¥ 0 | 17%

GTEx Â· median DERIVED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 11.62

> *Defined on: arms with a GTEx healthy anchor â€" âš MEASURED leg only; floor0 is NEVER emitted as a value*

---

hly_gtex_pct
ARM | PERCENT | 17%

GTEx Â· percent DERIVED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 0.9636

> *Defined on: arms with a GTEx healthy anchor â€" âš MEASURED leg only; floor0 is NEVER emitted as a value*

---

hly_nat_detect_n
ARM | COUNT | 91%

In how many of the ~104 matched NAT samples this arm is detected above the floor. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 104

---

hly_nat_median
ARM | REAL Â‰¥ 0 | 74%

Median NAT abundance (fill 74.0%). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 15.09

---

hly_tumor_prev
ARM | FRACTION 0â€"1 | 91%

Prevalence of the arm above the detection floor in tumour (median 0.01). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

DOSE_ Â· 3

supporting â€" read these when the core raises a question

> Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic.
>
> **3** columns Â· rung **arm** Â· SUBSET of the edge block Â· coverage **12%â€"12%** (median 12%) Â· **2** share the block description
>
> 2× real â‰¥ 0 Â· 1× constant

dose_comp_retention
ARM | REAL Â‰¥ 0 | 12%

retention DERIVED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 0.99

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

dose_confounded
ARM | CONSTANT | 12%

Is this arm's NATâ†'tumour dose a composition or proliferation artifact (either gated retention < 0.4)? AUTHORED

values: False (291)

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' False

---

dose_prolif_retention
ARM | REAL Â‰¥ 0 | 12%

retention DERIVED

example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

CTX_ARM_ Â· 2

supporting â€” read these when the core raises a question

> **2** columns Â· rung **arm** Â· the two ARM-RUNG members of ctx_, which the other cards do not carry Â· coverage **29%â€“29%** (median 29%)
> 1× boolean Â· 1× real â‰¥ 0

ctx_arm_abundant
ARM | BOOLEAN | 29%

Is `ctx_arm_dose >= log2(11)`, i.e. is this arm abundant enough to plausibly repress? **Median 0.00 â€” most arms are NOT.** AUTHORED

values: False (499) Â· True (213)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

ctx_arm_dose
ARM | REAL Â‰¥ 0 | 29%

This arm's median **log2(RPM+1)** across tumour samples (0.13 â€“ 17.92, median 1.08). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 14.8

ABUND_ Â· 5

supporting â€” read these when the core raises a question

> The arm's own abundance profile across the cohort â€” median, SD, IQR, detection fraction.
> **5** columns Â· rung **arm** Â· coverage **88%â€“91%** (median 91%) Â· **4** share the block description
> 3× real â‰¥ 0 Â· 1× fraction 0â€“1 Â· 1× categorical

abund_frac_detected
ARM | FRACTION 0Â€“1 | 91%

fraction DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

---

abund_iqr
ARM | REAL Â‰¥ 0 | 91%

interquartile range DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9309

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

---

abund_med
ARM | REAL Â‰¥ 0 | 91%

median DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 14.8

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

---

abund_sd
ARM | REAL Â‰¥ 0 | 88%

standard deviation DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.7103

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

---

abund_sd_tertile
ARM | CATEGORICAL | 88%

Dispersion tertile â€” **718 / 716 / 718 by construction**, so it carries no information beyond `abund_sd`'s rank. AUTHORED

values: high (718) Â· low (718) Â· mid (716)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' high

ATTR_ Â· 14

reference â€” context, provenance and rarely-quoted detail

> Attribution rollup over the arm's edges â€” how much of the gene's coupling this arm is credited with, and over how many reliable edges.
> **14** columns Â· rung **arm** Â· coverage **6%â€“29%** (median 15%) Â· **9** share the block description
> 11× fraction 0â€“1 Â· 2× count Â· 1× signed âˆ’1â€¦1

`attr_max_budget_share`
`ARM` `FRACTION 0Â€“1` `15%`

maximum Â· pressure budget Â· share `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.2604`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_max_credit_share`
`ARM` `FRACTION 0Â€“1` `6%`

maximum Â· attribution credit Â· share `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.8519`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_max_joint_share`
`ARM` `FRACTION 0Â€“1` `15%`

Largest JOINT attribution share among this arm's edges. `AUTHORED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.3167`

`attr_max_nested_share`
`ARM` `FRACTION 0Â€“1` `15%`

Largest NESTED attribution share. See `attr_max_joint_share`: different values (agree on 35.4%), one axis (spearman +0.962). `AUTHORED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.3167`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_max_within_share`
`ARM` `FRACTION 0Â€“1` `15%`

Largest WITHIN-cell attribution share for this arm. `AUTHORED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 1`

`attr_med_budget_share`
`ARM` `FRACTION 0Â€“1` `15%`

median Â· pressure budget Â· share `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1036`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_med_credit_share`
`ARM` `FRACTION 0Â€“1` `6%`

median Â· attribution credit Â· share `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.2493`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_med_dose_share`
`ARM` `FRACTION 0Â€“1` `29%`

Median dose share of this arm within its family cells. `AUTHORED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1582`

`attr_med_joint_share`
`ARM` `FRACTION 0Â€“1` `15%`

median Â· share `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.07153`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_med_nested_share`
`ARM` `FRACTION 0Â€“1` `15%`

median Â· share `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1393`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_med_phi_joint`
`ARM` `SIGNED Â^’1Â€¦1` `15%`

median `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.008336`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_med_within_share`
`ARM` `FRACTION 0Â€“1` `15%`

median Â· share `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.33`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_n_edges`
`ARM` `COUNT` `22%`

count `DERIVED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 23`

> *Defined on: arms with a Shapley/attribution decomposition over their edges (728)*

`attr_n_reliable`
`ARM` `COUNT` `15%`

Reliable edges behind this arm's `attr_*` shares (median 2). `AUTHORED`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 4`

GRANK_ Â· 3

reference â€” context, provenance and rarely-quoted detail

> The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from.
>
> `3 columns Â· rung arm Â· coverage 15%â€“20% (median 20%) Â· 3 share the block description`
> `3× count`

grank_HLY
ARM | COUNT | 15%

in the GTEx-healthy state DERIVED
example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 96`

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)*

---

grank_NAT
ARM | COUNT | 20%

in matched normal (NAT) DERIVED
example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 99`

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

---

grank_TUM
ARM | COUNT | 20%

in tumour DERIVED
example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†’ 99`

> *Defined on: edges present in the progression panel (arm expressed + state measured)*

REAL_ Â· 11

reference â€” context, provenance and rarely-quoted detail

> Realization rollup for the ARM â€” how many of its edges are scored, and its median and best realized coupling.
>
> `11 columns Â· rung arm Â· coverage 9%â€“24% (median 24%) Â· 8 share the block description`
> `7× count Â· 2× signed âˆ'1â€¦1 Â· 2× fraction 0â€“1`

real_best_coupling
ARM | SIGNED âˆ'1â€¦1 | 24%

Most negative realized coupling among this arm's scored genes (median âˆ'0.08, min âˆ'0.56). AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ -0.306`

---

real_frac_c10
ARM | FRACTION 0â€“1 | 9%

Fraction of this arm's scored genes coupling below âˆ'0.10. AUTHORED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.2609`

---

real_frac_sig
ARM | FRACTION 0â€“1 | 9%

fraction DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.2609`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

---

real_med_coupling
ARM | SIGNED âˆ'1â€¦1 | 24%

median DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.046`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

---

real_n_c05
ARM | COUNT | 24%

count Â· at the âˆ'0.05 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 6`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

---

real_n_c10
ARM | COUNT | 24%

count Â· at the âˆ'0.10 threshold DERIVED
example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 6`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

real_n_c15

ARM | COUNT | 24%

count Â· at the âˆ’0.15 threshold DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 6`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

---

real_n_c20

ARM | COUNT | 24%

count Â· at the âˆ’0.20 threshold DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 6`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

---

real_n_c30

ARM | COUNT | 24%

count Â· at the âˆ’0.30 threshold DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 1`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

---

real_n_scored

ARM | COUNT | 24%

Genes scored for this arm's realized coupling (median 3). âœ… Its `n_c05 â€¦ n_c30` ladder is properly nested â€” **0 violations** (verified). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 23`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

---

real_n_sig

ARM | COUNT | 24%

count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 6`

> *Defined on: arms with â‰¥1 calibrated-coupling-scored edge (593) â€” the realization LADDER; rates are gated at nâ‰¥5*

TS_ Â· 10

reference â€” context, provenance and rarely-quoted detail

> TargetScan context++ scores for the arm's sites â€” number of sites, mean and best context score. More negative context++ = stronger predicted repression.
>
> **10** columns Â· rung **arm** Â· coverage **13%â€“13%** (median 13%) Â· **8** share the block description
>
> 5× count Â· 2× real, signed Â· 1× signed âˆ’1â€¦1 Â· 1× fraction 0â€“1 Â· 1× real â‰¥ 0

ts_ctx_best

ARM | REAL, SIGNED | 13%

Best (most negative) TargetScan context++ score for this arm â€” range [âˆ’2.02, âˆ’0.34], **all negative by construction** (context++ scores repression as a negative number). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ -0.809`

---

ts_ctx_mean

ARM | SIGNED âˆ’1â€¦1 | 13%

mean DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ -0.283`

> *Defined on: arms in TargetScan's human default predictions (âš only 321)*

---

ts_frac_8mer

ARM | FRACTION 0â€“1 | 13%

fraction Â· 8mer sites DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.3189`

> *Defined on: arms in TargetScan's human default predictions (âš only 321)*

---

ts_kd_med

ARM | REAL, SIGNED | 13%

median DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ -5.181`

> *Defined on: arms in TargetScan's human default predictions (âš only 321)*

---

ts_n_7mer_A1

ARM | COUNT | 13%

count Â· 7mer sites DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 463`

> *Defined on: arms in TargetScan's human default predictions (âš only 321)*

---

ts_n_7mer_m8

ARM | COUNT | 13%

count Â· 7mer sites DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†’ 481`

> *Defined on: arms in TargetScan's human default predictions (âš only 321)*

---

ts_n_8mer
ARM COUNT 13%

count Â· 8mer sites DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 442

> *Defined on: arms in TargetScan's human default predictions (âš only 321)*

---

ts_n_genes
ARM COUNT AGG EDGE 13%

Genes with a TargetScan context++ site â€" median 497, fill **12.9%, the SPARSEST universe on the card**. Correlates +0.726 with the MANE seed scan and only **+0.231** with scanMiR-any. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1207

---

ts_n_sites
ARM COUNT AGG EDGE 13%

count DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1386

> *Defined on: arms in TargetScan's human default predictions (âš only 321)*

---

ts_sites_per_gene
ARM REAL â‰¥ 0 13%

per gene DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1.148

> *Defined on: arms in TargetScan's human default predictions (âš only 321)*

ISOC_ Â· 5

reference â€" context, provenance and rarely-quoted detail

> isomiR COMPOSITION â€" what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm.
> **5** columns Â· rung **arm** Â· coverage **91%â€"91%** (median 91%) Â· **4** share the block description
> 3× fraction 0â€"1 Â· 1× real â‰¥ 0 Â· 1× count

isoc_canonical_frac
ARM FRACTION 0â€"1 91%

canonical sites Â· fraction DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9956

> *Defined on: arms in the isomiR seed-COMPOSITION table â€" where the reads go; â isoc_orphan_frac flags MH-153 mass loss*

---

isoc_donated_frac
ARM FRACTION 0â€"1 91%

fraction DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.0029

> *Defined on: arms in the isomiR seed-COMPOSITION table â€" where the reads go; â isoc_orphan_frac flags MH-153 mass loss*

---

isoc_med_reads
ARM REAL â‰¥ 0 91%

Median isomiR reads for this arm â€" **median 1.00, max 528,927**. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 2.229e+04

---

isoc_n_dest_families
ARM COUNT 91%

count DERIVED

values: 0.0 (1,283) Â· 1.0 (623) Â· 2.0 (238) Â· 3.0 (59) Â· 4.0 (13) Â· 5.0 (3) Â· 6.0 (3)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

> *Defined on: arms in the isomiR seed-COMPOSITION table â€" where the reads go; â isoc_orphan_frac flags MH-153 mass loss*

---

isoc_orphan_frac
ARM FRACTION 0â€"1 91%

fraction DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.0034

> *Defined on: arms in the isomiR seed-COMPOSITION table â€" where the reads go; â isoc_orphan_frac flags MH-153 mass loss*

SDSZ_ Â· 4

reference â€” context, provenance and rarely-quoted detail

MANE 3'UTR seed scan.

4 columns Â· rung **arm** Â· coverage **97%â€“97%** (median 97%) Â· **2** share the block description
2× count Â· 1× real â‰¥ 0 Â· 1× fraction 0â€“1

sdsz_N_7mer_plus
ARM | COUNT | AGG EDGE | 97%

count Â· 7mer sites DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 3013

> *Defined on: arms in the MANE 3'UTR seed scan â€" âš a FOURTH targetome universe, never blend (2,370)*

sdsz_N_8mer
ARM | COUNT | AGG EDGE | 97%

count Â· 8mer sites DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 713

> *Defined on: arms in the MANE 3'UTR seed scan â€" âš a FOURTH targetome universe, never blend (2,370)*

sdsz_N_eff
ARM | REAL Â‰¥ 0 | AGG EDGE | 97%

Effective seed-site count over the scanned universe (median 2,094, fill **96.7%** â€" the best-covered targetome axis). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1863

sdsz_rarity
ARM | FRACTION 0â€"1 | 97%

The arm's seed RARITY, min-max normalised to 0â€"1 (median 0.11). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1253

CHIM_ Â· 5

reference â€” context, provenance and rarely-quoted detail

Chimeric-evidence ROLLUP for the arm (aggregated over its edges).

5 columns Â· rung **arm** Â· coverage **21%â€“63%** (median 60%) Â· **3** share the block description
4× count Â· 1× real â‰¥ 0

chim_manakov_n_genes
ARM | COUNT | AGG EDGE | 60%

count Â· genes DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 7297

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_manakov_weight
ARM | REAL Â‰¥ 0 | AGG EDGE | 60%

Manakov chimeric support summed over this arm's edges (median 6, max 210,722). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 9900

chim_n_sources
ARM | COUNT | AGG EDGE | 63%

count DERIVED
values: 1.0 (1,079) Â· 2.0 (229) Â· 4.0 (128) Â· 3.0 (102)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 4

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_tarbase_n_genes
ARM | COUNT | AGG EDGE | 21%

count Â· genes DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 2701

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

chim_tarbase_weight

ARM · COUNT · AGG EDGE · 21%

TarBase chimeric support for the arm (median 8, max 10,255), defined on 20.9% of arms â€" a third of the Manakov coverage. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1.026e+04`

> *Defined on: arms with chimeric ligation evidence (1,538) â€" âš ~90% Manakov/HEK293T, per-source, NEVER pooled*

FST_ Â· 16

reference â€" context, provenance and rarely-quoted detail

> â THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES â€" gene-free (`card_ladders.family_state_shift` , 2026-08-19).
>
> **16** columns Â· rung **arm** Â· coverage **3%â€"100%** (median 20%) Â· **7** share the block description
> 6× count Â· 5× boolean Â· 3× fraction 0â€"1 Â· 2× signed âˆ'1â€¦1

fst_d_share_HLY_TUM

ARM · SIGNED âˆ'1â€¦1 · 3%

As `fst_d_share_NAT_TUM` but GTEx-healthy â†' tumour, same common-support correction. n=68 (HLY is the sparsest leg at 17.0%). Post-fix **mean âˆ'0.0000, 47.1% positive, p=0.93.** AUTHORED

example: `hsa-miR-26b-5p / miR-26-5p â†' 0.2402`

---

fst_d_share_NAT_TUM

ARM · SIGNED âˆ'1â€¦1 · 6%

Change in the arm's share of its seed family's linear dose, matched NAT â†' tumour, **re-normalised over the members measured in BOTH states.** AUTHORED

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' -0.0072`

---

fst_dominance_switch

ARM · BOOLEAN · 7%

Did the family's DOMINANT arm change between healthy and tumour (103 true / 74 false)? AUTHORED

values: `1.0 (103) Â· 0.0 (74)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 0`

---

fst_hly_family_measured

ARM · BOOLEAN · 100%

Do ALL members of this arm's family have a measured GTEx level. AUTHORED

values: `False (2,142) Â· True (308)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' False`

---

fst_hly_measured

ARM · BOOLEAN · 100%

in the GTEx-healthy state DERIVED

values: `False (2,033) Â· True (417)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' False`

---

fst_is_dominant_HLY

ARM · BOOLEAN · 17%

the dominant one Â· in the GTEx-healthy state DERIVED

values: `True (345) Â· False (72)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' True`

---

fst_is_dominant_TUM

ARM · BOOLEAN · 20%

the dominant one Â· in tumour DERIVED

values: `True (394) Â· False (91)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' True`

---

fst_n_common_HLY_TUM

ARM · COUNT · 9%

Members measured in BOTH GTEx-healthy and tumour â€" the denominator behind `fst_d_share_HLY_TUM` . See `fst_n_common_NAT_TUM` . AUTHORED

values: `1.0 (162) Â· 2.0 (48) Â· 4.0 (8) Â· 6.0 (6) Â· 3.0 (6)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1`

---

fst_n_common_NAT_TUM

ARM · COUNT · 20%

How many of the family's arms are measured in BOTH NAT and tumour â€" the DENOMINATOR behind `fst_d_share_NAT_TUM` , shipped because the column was wrong for want of it. AUTHORED

values: `1.0 (343) Â· 2.0 (86) Â· 4.0 (16) Â· 3.0 (12) Â· 8.0 (8) Â· 7.0 (7) Â· 6.0 (6)`

example: `hsa-miR-200c-3p / miR-200bc-3p/429 â†' 2`

---

fst_n_members
ARM COUNT 100%

count Â· family members DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 10

fst_rank_HLY
ARM COUNT 17%

rank Â· in the GTEx-healthy state DERIVED
values: 1.0 (348) Â· 2.0 (47) Â· 3.0 (13) Â· 4.0 (6) Â· 5.0 (2) Â· 6.0 (1)
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1

fst_rank_NAT
ARM COUNT 74%

rank Â· in matched normal (NAT) DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

fst_rank_TUM
ARM COUNT 20%

rank Â· in tumour DERIVED
values: 1.0 (412) Â· 2.0 (46) Â· 3.0 (11) Â· 4.0 (9) Â· 7.0 (2) Â· 5.0 (2) Â· 6.0 (2) Â· 8.0 (1)
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1

fst_share_HLY
ARM FRACTION 0â€“1 17%

Share of family dose in GTEx-healthy, from `hly_gtex_median` (fill **17.0%** â€" the sparsest leg). Same singleton degeneracy; and cross-platform, so contrast only. AUTHORED
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 0.9533

fst_share_NAT
ARM FRACTION 0â€“1 74%

Share of family dose in matched NAT, from `hly_nat_median` (fill **74.0%** â€" the best-covered leg, 3.7Ã— TUM). Same singleton degeneracy: median 1.000, mask on `fst_n_members > 1`. AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.4124

fst_share_TUM
ARM FRACTION 0â€“1 20%

The arm's fraction of its seed family's total linear dose in tumour, from `arm_med_rpm` (fill **19.8%**). AUTHORED
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 0.945

DGLOBAL_ Â· 3

reference â€" context, provenance and rarely-quoted detail

> Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step.
>
> **3** columns Â· rung **arm** Â· coverage **15%â€“20%** (median 15%) Â· **3** share the block description
> 3× real, signed

dGlobal_HLY_NAT
ARM REAL, SIGNED 15%

in the GTEx-healthy state Â· in matched normal (NAT) DERIVED
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 2

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

dGlobal_HLY_TUM
ARM REAL, SIGNED 15%

in the GTEx-healthy state Â· in tumour DERIVED
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 3

> *Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€" many arms are multi-mapping-collapsed, MH-166)*

dGlobal_NAT_TUM
ARM REAL, SIGNED 20%

in matched normal (NAT) Â· in tumour DERIVED
example: hsa-miR-200c-3p / miR-200bc-3p/429 â†' 1

> *Defined on: edges with a matched NAT leg (n~104 paired participants)*

SEQ_ Â· 4

reference â€" context, provenance and rarely-quoted detail

> **4** columns Â· rung **arm** Â· coverage **30%â€“30%** (median 30%)
> 2× count Â· 2× real â‰¥ 0

seq_n_genes_any
ARM | COUNT | AGG EDGE | 30%

Genes with ANY scanMiR K_D site â€" AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1.124e+04

seq_n_genes_strong
ARM | COUNT | AGG EDGE | 30%

Genes with a STRONG scanMiR K_D site â€" median 3,632, spearman **+0.822** with `sdsz_N_eff` and **+0.797** with `site_n_genes_canonical`, but only **+0.566** with `ts_n_genes`. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 4550

seq_promisc
ARM | REAL Â‰¥ 0 | 30%

log-scale promiscuity over STRONG scanMiR sites (4.45 â€" 8.94). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 8.423

seq_promisc_any
ARM | REAL Â‰¥ 0 | 30%

Promiscuity over ANY scanMiR site (6.46 â€" 9.45) â€" the permissive cut. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 9.327

SUB_ Â· 6

reference â€" context, provenance and rarely-quoted detail

> PAM50 subtype block aggregated to the ARM. See `esub_` for the per-edge form.
> **6** columns Â· rung **arm** Â· coverage **30%â€"30%** (median 30%) Â· **4** share the block description
> 2× count Â· 2× signed âˆ'1â€¦1 Â· 1× categorical Â· 1× fraction 0â€"1

sub_best_frac
ARM | FRACTION 0â€"1 | 30%

the best Â· fraction DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.3684

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

sub_best_mode
ARM | CATEGORICAL | 30%

the best DERIVED

values: Basal (278) Â· Her2 (240) Â· LumB (105) Â· LumA (105)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' Her2

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

sub_med_rho_best
ARM | SIGNED âˆ'1â€¦1 | 30%

Median coupling in the arm's BEST subtype. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' -0.03409

sub_med_rho_pool
ARM | SIGNED âˆ'1â€¦1 | 30%

Median pooled-subtype coupling (median âˆ'0.0064) â€" the unselected reference for `sub_med_rho_best`. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.06516

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

sub_n_edges
ARM | COUNT | 30%

count DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 38

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

sub_n_q05
ARM | COUNT | 30%

count DERIVED

values: 0.0 (659) Â· 1.0 (54) Â· 2.0 (10) Â· 3.0 (3) Â· 5.0 (2)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0

> *Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) â€" âš DISTRIBUTIONAL (MH-165), not a validated label*

cov_ Â· 22

reference â€" context, provenance and rarely-quoted detail

> Coverage flags â€" was this row ever SCANNED by the named source.
>
> **22** columns Â· rung **arm** Â· coverage **100%â€"100%** (median 100%) Â· **2** share the block description
>
> 22× boolean

cov_ago_dom
ARM BOOLEAN 100%

Was this arm ever SCANNED for AGO-loading dominance from Manakov CLIP? **True on 678 of 2,450 arms.** AUTHORED

values: False (1,772) Â· True (678)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_attr
ARM BOOLEAN 100%

Was this arm ever SCANNED for the Shapley/NNLS attribution block? **True on 546 of 2,450 arms.** AUTHORED

values: False (1,904) Â· True (546)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_beta
ARM BOOLEAN 100%

Î² DERIVED

values: False (1,722) Â· True (728)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_cnv
ARM BOOLEAN 100%

Was this arm ever SCANNED for the arm-locus copy-number block? **True on 856 of 2,450 arms.** AUTHORED

values: False (1,594) Â· True (856)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_comp
ARM BOOLEAN 100%

Was this arm ever SCANNED for the compartment-loading (`cload_`) block? **True on 205 of 2,450 arms.** AUTHORED

values: False (2,245) Â· True (205)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_cptac
ARM BOOLEAN 100%

CPTAC DERIVED

values: True (2,071) Â· False (379)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

> *Defined on: arms measurable in CPTAC (2,095)*

---

cov_curated
ARM BOOLEAN 100%

Was this arm ever SCANNED for the curated-literature blocks (`fame_`, `cur_`)? **True on 875 of 2,450 arms.** AUTHORED

values: False (1,575) Â· True (875)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_expr
ARM BOOLEAN 100%

Was this arm ever SCANNED for the expression / abundance blocks? **True on 2,231 of 2,450 arms.** AUTHORED

values: True (2,231) Â· False (219)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_famrole
ARM BOOLEAN 100%

Was this arm ever SCANNED for the family-role block? **True on 2,231 of 2,450 arms.** AUTHORED

values: True (2,231) Â· False (219)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_field
ARM BOOLEAN 100%

Was this arm ever SCANNED for the focal-field correlation block? **True on 571 of 2,450 arms.** AUTHORED

values: False (1,879) Â· True (571)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

cov_focal

ARM BOOLEAN 100%

Was this arm ever SCANNED for the focal-locus CNV block? **True on 733 of 2,450 arms.** AUTHORED

values: False (1,717) Â· True (733)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_fst

ARM BOOLEAN 100%

Was the family-share block computed for this arm at all â€" i.e. did it have the abundance input. AUTHORED

values: False (1,965) Â· True (485)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' False

---

cov_gctx

ARM BOOLEAN 100%

Was this arm ever SCANNED for the genomic-context block? **True on 714 of 2,450 arms.** AUTHORED

values: False (1,736) Â· True (714)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_isocomp

ARM BOOLEAN 100%

Was this arm ever SCANNED for the isomiR-composition block? **True on 2,222 of 2,450 arms.** AUTHORED

values: True (2,222) Â· False (228)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_isomir

ARM BOOLEAN 100%

Was this arm ever SCANNED for the isomiR seed-shift block? **True on 687 of 2,450 arms.** AUTHORED

values: False (1,763) Â· True (687)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_meth

ARM BOOLEAN 100%

Was this arm ever SCANNED for the locus-methylation (`bc_meth_`) block? **True on 343 of 2,450 arms.** AUTHORED

values: False (2,107) Â· True (343)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_real

ARM BOOLEAN 100%

Was this arm ever SCANNED for the realized-coupling (`real_`) block? **True on 590 of 2,450 arms.** AUTHORED

values: False (1,860) Â· True (590)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_seed_scan

ARM BOOLEAN 100%

Was this arm ever SCANNED for the MANE 3'UTR seed scan (`site_`)? AUTHORED

values: True (2,370) Â· False (80)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_seq_scan

ARM BOOLEAN 100%

Was this arm ever SCANNED for the scanMiR K_D scan (`seq_`)? AUTHORED

values: False (1,714) Â· True (736)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_site_types

ARM BOOLEAN 100%

Was this arm ever SCANNED for the site-type breakdown? AUTHORED

values: False (1,732) Â· True (718)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_subtype

ARM BOOLEAN 100%

Was this arm ever SCANNED for the PAM50 subtype block (`sub_`)? AUTHORED

values: False (1,722) Â· True (728)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

cov_targetscan

ARM BOOLEAN 100%

Was this arm ever SCANNED for the TargetScan context++ block (`ts_`)? AUTHORED

values: False (2,133) Â· True (317)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

CUR_HE_ Â· 2

reference â€” context, provenance and rarely-quoted detail

> **2** columns Â· rung **arm** Â· coverage **36%â€“36%** (median 36%)
> 2× count

cur_he_degree
ARM | COUNT | AGG EDGE | 36%

How many HE genes this arm regulates in the curated design â€” the arm's curation degree (median 3, max 125). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 51

> *Defined on: arms with miRTarBase curation / ledger PMIDs â€" â›" FAME axes, not targetome (MH-208)*

cur_he_degree_expr
ARM | COUNT | AGG EDGE | 36%

As `cur_he_degree` but counting only targets above the expression floor. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 43

ISO_ Â· 5

reference â€” context, provenance and rarely-quoted detail

> isomiR profile for the arm â€" 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X_fam, which is BUILT but default-OFF (coupling wash, bidirectional).
> **5** columns Â· rung **arm** Â· coverage **28%â€“28%** (median 28%) Â· **4** share the block description
> 2× fraction 0â€“1 Â· 1× count Â· 1× real â‰¥ 0 Â· 1× real, signed

iso_n_samples
ARM | COUNT | 28%

count DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1078

> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

iso_rpm_summed
ARM | REAL â‰¥ 0 | 28%

âš **A SUM ACROSS SAMPLES, not a per-sample RPM â€" the name misleads.** Max **270,988,199**, which is impossible for a per-million unit; divided by `iso_n_samples` the scale is sane (median **57**, max 251,381). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 3.553e+07

iso_seed_shift_frac
ARM | FRACTION 0â€“1 | 28%

shift Â· fraction DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.004

> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

iso_seed_shift_sd
ARM | FRACTION 0â€“1 | 28%

shift Â· standard deviation DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.004

> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

iso_top_shift_nt
ARM | REAL, SIGNED | 28%

the top-ranked Â· shift DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 3

> *Defined on: arms with a 5'-isomiR seed-shift measurement (687) â€" âš gate on iso_n_samples, the median is n-DEPENDENT*

FIELD_ Â· 5

reference â€” context, provenance and rarely-quoted detail

> Within-patient NAT field-effect fit â€" is the regulator already established in the patient's adjacent normal before the tumour. Defined on the paired subset.
> **5** columns Â· rung **arm** Â· coverage **23%â€“23%** (median 23%) Â· **3** share the block description
> 4× signed âˆ'1â€¦1 Â· 1× count

field_excess_over_perm

ARM | SIGNED Â˘'1Â€¦1 | 23%

â›"â›" **NOT A RETENTION RATIO â€" a FIFTH sense of the word in this codebase.** Measured: it equals `field_r_own â˘' field_r_perm` (exact within rounding on 100% of 571 arms, max residual 0.001), i.e. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.363`

---

field_n

ARM | COUNT | 23%

count DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 103`

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

---

field_r_adj

ARM | SIGNED Â˘'1Â€¦1 | 23%

correlation Â· composition-adjusted DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.367`

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

---

field_r_own

ARM | SIGNED Â˘'1Â€¦1 | 23%

correlation DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.343`

> *Defined on: arms with a within-patient NAT field-effect fit (571)*

---

field_r_perm

ARM | SIGNED Â˘'1Â€¦1 | 23%

The permutation null for `field_r_own` â€" median **0.000**, range [â˘'0.11, +0.09]. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.02`

FAMROLE_ Â· 6

reference â€" context, provenance and rarely-quoted detail

> The arm's ROLE inside its seed family â€" member count, its share of family abundance, and its rank among members.
>
> **6** columns Â· rung **arm** Â· coverage **88%â€"100%** (median 91%) Â· **5** share the block description
>
> 3× count Â· 1× fraction 0â€"1 Â· 1× categorical Â· 1× boolean

famrole_abund_rank

ARM | COUNT | 91%

unweighted abundance-sum reference Â· rank DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1`

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

---

famrole_abund_share

ARM | FRACTION 0â€"1 | 91%

unweighted abundance-sum reference Â· share DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.465`

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

---

famrole_class

ARM | CATEGORICAL | 100%

The arm's role in its seed family â€" **`sole` on 1,639 of 2,450 (66.9%)**, `dominant` 282, `co` and the rest below. AUTHORED

values: `sole (1,639) Â· dominant (282) Â· co (267) Â· minor (262)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' co`

---

famrole_is_dominant

ARM | BOOLEAN | 91%

the dominant one DERIVED

values: `True (1,779) Â· False (452)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' False`

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

---

famrole_n_members

ARM | COUNT | 100%

count Â· family members DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 10`

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

famrole_var_rank
ARM | COUNT | 88%

rank DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 7

> *Defined on: arms with a seed family â€" the arm's ROLE inside it; â›" DEGENERATE at famrole_n_members==1*

BC_ Â· 11

reference â€" context, provenance and rarely-quoted detail

Breast-cancer locus annotation for the arm â€" CpG probes, CGI overlap and related epigenetic context at the precursor locus.
11 columns Â· rung **arm** Â· coverage **14%â€"36%** (median 14%) Â· **9** share the block description
4× fraction 0â€"1 Â· 4× count Â· 1× boolean Â· 1× signed âˆ'1â€¦1 Â· 1× categorical

bc_fam_context_homogeneous
ARM | BOOLEAN | 36%

per seed family DERIVED
values: True (720) Â· False (160)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' False

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_fam_dose_coding_host_frac
ARM | FRACTION 0â€"1 | 36%

per seed family Â· dose Â· fraction DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.121

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_fam_dose_cotranscribed_frac
ARM | FRACTION 0â€"1 | 36%

per seed family Â· dose Â· fraction DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.105

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_fam_n_distinct_coding_hosts
ARM | COUNT | 36%

per seed family Â· count DERIVED
values: 1.0 (424) Â· 0.0 (379) Â· 2.0 (45) Â· 3.0 (32)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 3

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_fam_n_members
ARM | COUNT | 36%

per seed family Â· count Â· family members DERIVED
values: 1.0 (606) Â· 2.0 (139) Â· 3.0 (69) Â· 5.0 (22) Â· 10.0 (21) Â· 8.0 (8) Â· 4.0 (8) Â· 6.0 (7)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 10

> *Defined on: âš BROADCAST from the seed FAMILY (genomic_context.family_context)*

---

bc_meth_dbeta
ARM | SIGNED âˆ'1â€¦1 | 14%

Tumour-minus-normal methylation beta at this arm's locus (median âˆ'0.03, range âˆ'0.32â€¦+0.23). AUTHORED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1451

---

bc_meth_direction
ARM | CATEGORICAL | 14%

Tumour-vs-normal methylation direction at this arm's locus. AUTHORED
values: none (287) Â· hypo (46) Â· hyper (10)
example: hsa-let-7a-5p / let-7-5p/98-5p â†' none

---

bc_meth_n_cgi
ARM | COUNT | 14%

count DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€" 5p and 3p share one promoter and one Î"Î² (354 arms)*

---

bc_meth_n_probes
ARM | COUNT | 14%

count DERIVED
example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€" 5p and 3p share one promoter and one Î"Î² (354 arms)*

---

bc_meth_normal_beta

ARM | FRACTION 0â€“1 | 14%

Î² DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.6494

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€” 5p and 3p share one promoter and one Î”Î² (354 arms)*

bc_meth_tumour_beta

ARM | FRACTION 0â€“1 | 14%

Î² DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.7945

> *Defined on: âš BROADCAST from the pri-miRNA HAIRPIN â€” 5p and 3p share one promoter and one Î”Î² (354 arms)*

GCTX_ Â· 16

reference â€” context, provenance and rarely-quoted detail

> Genomic context of the arm's precursor locus (`genomic_context.classify_he_arms`): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware.
>
> **16** columns Â· rung **arm** Â· coverage **24%â€“29%** (median 29%) Â· **6** share the block description
>
> 6× categorical Â· 4× boolean Â· 3× identifier Â· 1× constant Â· 1× count Â· 1× real â‰¥ 0

gctx_clustered

ARM | BOOLEAN | 29%

Does this arm's precursor sit in a genomic CLUSTER with other miRNAs (310 true / 404 false)? AUTHORED

values: False (404) Â· True (310)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True

gctx_context_mixed

ARM | BOOLEAN | 29%

Do this arm's multiple loci DISAGREE about their genomic context â€” **20 arms of 714**? AUTHORED

values: False (694) Â· True (20)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ False

gctx_coord_source

ARM | CONSTANT | 29%

Provenance of the genomic-context coordinates. AUTHORED

values: gencode (714)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ gencode

gctx_host

ARM | IDENTIFIER | 27%

The HOST GENE the precursor sits inside (356 distinct; `ENSG00000269842` , `MEG9` , `MEG3` most common). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ MIR100HG

gctx_host_region

ARM | IDENTIFIER | 27%

WHERE in the host the precursor sits (74 distinct; `intron 2` and `intron 1` most common, 61 arms each). An intronic miRNA is spliced from its host's pre-mRNA; an exonic one competes with it. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ exon 41

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€” 714)*

gctx_host_type

ARM | CATEGORICAL | 27%

The host's BIOTYPE â€” `protein_coding` 364 Â· `lncRNA` 303, plus 5 rarer levels. AUTHORED

values: protein_coding (364) Â· lncRNA (303) Â· transcribed_unprocessed_pseudogene (2) Â· TEC (1) Â· transcribed_unitary_pseudogene (1) Â· processed_pseudogene (1) Â· IG_V_gene (1)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ lncRNA

gctx_in_exon

ARM | BOOLEAN | 29%

Is the precursor inside a host EXON (218 true / 496 false) rather than an intron? AUTHORED

values: False (496) Â· True (218)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True

gctx_mir_class
ARM · CATEGORICAL · 29%

class label DERIVED

values: sense_coding_host (337) Â· sense_lncRNA_host (296) Â· intergenic (41) Â· antisense_overlap (40)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' sense_lncRNA_host

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_mirgenedb
ARM · BOOLEAN · 29%

Is this arm in **MirGeneDB** â€" the curated high-confidence miRNA set (526 true / 188 false)? AUTHORED

values: True (526) Â· False (188)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' True

---

gctx_n_loci
ARM · COUNT · 29%

count DERIVED

values: 1.0 (642) Â· 2.0 (62) Â· 3.0 (10)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 3

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_class
ARM · CATEGORICAL · 29%

class label DERIVED

values: host_shared (346) Â· own (239) Â· shared_other (128) Â· unknown (1)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' host_shared

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_coord_class
ARM · CATEGORICAL · 29%

class label DERIVED

values: host_shared (480) Â· shared_other (111) Â· unknown (85) Â· intergenic (38)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' host_shared

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_gene
ARM · IDENTIFIER · 24%

per gene DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' MIR100HG

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_gene_type
ARM · CATEGORICAL · 24%

per gene DERIVED

values: protein_coding (320) Â· lncRNA (267) Â· misc_RNA (2) Â· transcribed_processed_pseudogene (1) Â· IG_V_gene (1)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' lncRNA

> *Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms â€" 714)*

---

gctx_promoter_host_tss_kb
ARM · REAL â‰¥ 0 · 24%

Distance in kb from the precursor to its host's TSS (median **0.5 kb**, max 874). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 376.1

---

gctx_strand_rel
ARM · CATEGORICAL · 27%

Is the precursor SENSE (633 arms) or ANTISENSE (40) to its host? AUTHORED

values: sense (633) Â· antisense (40)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' sense

SURV_ Â· 8

reference â€" context, provenance and rarely-quoted detail

> TCGA outcome panel legs.
>
> **8** columns Â· rung **arm** Â· coverage **6%â€"6%** (median 6%) Â· **5** share the block description
>
> 4× p-value Â· 3× real â‰¥ 0 Â· 1× constant

surv_dfi_hr_adj
ARM · REAL â‰¥ 0 · 6%

Adjusted DFI hazard ratio per arm â€" range **[0.84, 1.16]**, median **1.00**. The whole distribution sits within Â±16% of no effect. See `surv_dfi_q_adj`. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9371

> *Defined on: arms in the TCGA outcome panel (145) â€" â†' TCGA legs only; the GSE leg is bare-stem matched*

surv_dfi_hr_base
ARM | REAL Â‰¥ 0 | 6%

Unadjusted DFI hazard ratio for the arm (0.86 â€“ 1.16, median 1.00). AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9463`

---

surv_dfi_p_adj
ARM | P-VALUE | 6%

p-value Â· composition-adjusted DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.4701`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

---

surv_dfi_p_base
ARM | P-VALUE | 6%

p-value DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.5368`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

---

surv_dfi_q_adj
ARM | CONSTANT | 6%

BH q for the arm's DFI hazard ratio. AUTHORED

values: `0.996528491082692 (145)`

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9965`

---

surv_os_hr_adj
ARM | REAL Â‰¥ 0 | 6%

composition-adjusted DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 1.011`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

---

surv_os_p_adj
ARM | P-VALUE | 6%

p-value Â· composition-adjusted DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.8856`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

---

surv_os_q_adj
ARM | P-VALUE | 6%

BH-adjusted q Â· composition-adjusted DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9745`

> *Defined on: arms in the TCGA outcome panel (145) â€" â›" TCGA legs only; the GSE leg is bare-stem matched*

AID_ Â· 4

reference â€" context, provenance and rarely-quoted detail

> Arm-IDENTIFIABILITY rollup for the arm: how often this arm is separable from its same-seed mates (`arm_resolvable`, `arm_sep_z`, `oof_drho` medians).
>
> **4** columns Â· rung **arm** Â· coverage **6%â€“6%** (median 6%) Â· **3** share the block description
>
> 1× fraction 0â€“1 Â· 1× real â‰¥ 0 Â· 1× signed âˆ'1â€¦1 Â· 1× count

aid_frac_resolvable
ARM | FRACTION 0â€“1 | AGG ARM-IN-FAMILY | 6%

fraction Â· resolvability DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.4118`

> *Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€" the arm rung's 20.3%-of-edges footprint (142)*

---

aid_med_arm_sep_z
ARM | REAL Â‰¥ 0 | AGG ARM-IN-FAMILY | 6%

median Â· per arm Â· separation Â· z-score DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 2.381`

> *Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€" the arm rung's 20.3%-of-edges footprint (142)*

---

aid_med_oof_drho
ARM | SIGNED Âˆ'1Â€¦1 | AGG ARM-IN-FAMILY | 6%

median Â· out-of-fold DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' -0.004584`

> *Defined on: arms inside â‰¥1 MULTI-ARM (gene,family) cell â€" the arm rung's 20.3%-of-edges footprint (142)*

aid_n_cells

ARM COUNT

AGG ARM-IN-FAMILY 6%

Multi-arm cells this arm participates in â€” fill **5.8%** (the arm identifiability block is the sparsest on the card). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 17

WSHIFT_ Â· 6

reference â€" context, provenance and rarely-quoted detail

> Within-patient paired quantities on the ~103 matched tumour/NAT participants.
>
> **6** columns Â· rung **arm** Â· coverage **45%â€"91%** (median 52%) Â· **6** share the block description
>
> 3× real â‰¥ 0 Â· 2× real, signed Â· 1× count

wshift_mean_dGlobalRank

ARM REAL, SIGNED 52%

mean DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' -0.13

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

wshift_mean_own_shift

ARM REAL, SIGNED 52%

mean Â· shift DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' -0.223

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

wshift_n_pairs

ARM COUNT 91%

count DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 103

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

wshift_nat_patient_sd

ARM REAL â‰¥ 0 74%

in matched normal (NAT) Â· standard deviation DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.556

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

wshift_own_specific_frac

ARM REAL â‰¥ 0 45%

fraction DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.62

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

wshift_sd_dGlobalRank

ARM REAL â‰¥ 0 45%

standard deviation DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.4

> *Defined on: arms in the WITHIN-PATIENT paired subset (nâ‰ˆ103 matched tumour/NAT pairs; 2,236 arms)*

TIER_ Â· 5

reference â€" context, provenance and rarely-quoted detail

> Expression tier of the arm against the detection floor â€" class, fraction of samples expressed, p90 and max log2 RPM.
>
> **5** columns Â· rung **arm** Â· coverage **91%â€"91%** (median 91%) Â· **2** share the block description
>
> 2× real â‰¥ 0 Â· 1× categorical Â· 1× boolean Â· 1× fraction 0â€"1

tier_class

ARM CATEGORICAL 91%

Expression tier â€" `silent` **on 1,518 of 2,236 arms (67.9%)**, `conditional` 476, the rest robust. AUTHORED

values: silent (1,518) Â· conditional (476) Â· robust (242)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' robust

tier_expressed
ARM | BOOLEAN | 91%

Is the arm above the expression floor often enough to be usable (718 true / 1,518 false)? AUTHORED

values: False (1,518) Â· True (718)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True

---

tier_frac_expressed
ARM | FRACTION 0â€“1 | 91%

fraction DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

---

tier_max
ARM | REAL â‰¥ 0 | 91%

maximum DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 17.48

> *Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)*

---

tier_p90
ARM | REAL â‰¥ 0 | 91%

The arm's 90th-percentile RPM across tumour samples (0.10 â€“ 18.49, median 0.94). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 15.76

SURROGATE_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **0%â€“0%** (median 0%)
1× fraction 0â€“1 Â· 1× categorical

surrogate_corr
ARM | FRACTION 0â€“1 | 0%

Correlation between this arm and its surrogate (0.62 â€“ 0.89) â€” the surrogate's quality gate. See the edge twin. AUTHORED

values: 0.621 (1) Â· 0.874 (1) Â· 0.892 (1) Â· 0.842 (1)

example: hsa-miR-181a-5p / miR-181-5p â†’ 0.892

> *Defined on: arms with a same-seed SURROGATE (MH-166) â€” 3.6% of edges*

---

surrogate_instrument
ARM | CATEGORICAL | 0%

Which surrogate stands in for this arm's collapsed GTEx leg (4 levels). AUTHORED

values: hsa-miR-93-5p (1) Â· hsa-miR-20a-5p (1) Â· hsa-miR-181b-5p (1) Â· hsa-miR-429 (1)

example: hsa-miR-181a-5p / miR-181-5p â†’ hsa-miR-181b-5p

DISC_ Â· 2

reference â€” context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **1%â€“100%** (median 50%)
1× categorical Â· 1× count

disc_gold_families
ARM | CATEGORICAL | 1%

Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed). AUTHORED

example: hsa-miR-25-3p / miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ miR-25-3p/32-5p/92-3p/363-3p/367-3p

---

disc_n_gold_edges
ARM | COUNT | 100%

How many of the 157 discovery-queue edges name this arm. Concentrated: miR-106b-5p **39** Â· miR-20a-5p 25 Â· miR-93-5p 25 Â· miR-19a-3p 19 Â· miR-18a-5p 9. AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0

FOC_ Â· 7

reference â€” context, provenance and rarely-quoted detail

> Paralog-decontaminated FOCAL-locus concordance.
>
> **7** columns Â· rung **arm** Â· coverage **30%â€“30%** (median 30%) Â· **5** share the block description
>
> 2× count Â· 2× signed âˆ’1â€¦1 Â· 1× identifier Â· 1× boolean Â· 1× p-value

foc_focal_locus
ARM IDENTIFIER 30%

The focal MIRBase locus used for this arm's CN analysis (437 distinct). AUTHORED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ MI0000062

foc_multilocus
ARM BOOLEAN 30%

Does this arm map to MORE THAN ONE genomic locus (35 true / 698 false)? AUTHORED

values: False (698) Â· True (35)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ True

foc_n
ARM COUNT 30%

count DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 1032

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â›” DESCRIPTIVE, not an instrument*

foc_n_loci
ARM COUNT 30%

count DERIVED

values: 1.0 (698) Â· 2.0 (27) Â· 3.0 (8)

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 3

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â›” DESCRIPTIVE, not an instrument*

foc_partial_p
ARM P-VALUE 30%

p-value DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 6.541e-09

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â›” DESCRIPTIVE, not an instrument*

foc_partial_rho
ARM SIGNED âˆ’1â€¦1 30%

Spearman Ï DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.1871

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â›” DESCRIPTIVE, not an instrument*

foc_spearman_rho
ARM SIGNED âˆ’1â€¦1 30%

Spearman Ï DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 0.08216

> *Defined on: arms with a paralog-decontaminated FOCAL-locus concordance (733) â€” â›” DESCRIPTIVE, not an instrument*

CNV_ Â· 9

reference â€” context, provenance and rarely-quoted detail

> Copy number at the arm's own locus â€” mean CN and amplification/deletion summaries.
>
> **9** columns Â· rung **arm** Â· coverage **35%â€“35%** (median 35%) Â· **8** share the block description
>
> 6× percent Â· 3× real â‰¥ 0

cnv_mean_cn
ARM REAL â‰¥ 0 35%

mean DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†’ 2.403

> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€” âš multi-locus arms are DILUTED here*

cnv_median_cn
ARM | REAL Â‰¥ 0 | 35%

median DERIVED

values: 2.0 (632) Â· 3.0 (175) Â· 4.0 (40) Â· 2.5 (8) Â· 3.5 (1)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 2

> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€" âš multi-locus arms are DILUTED here*

---

cnv_pct_altered
ARM | PERCENT | 35%

percent DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 43.66

> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€" âš multi-locus arms are DILUTED here*

---

cnv_pct_amp
ARM | PERCENT | 35%

percent DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 16.21

> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€" âš multi-locus arms are DILUTED here*

---

cnv_pct_deep_del
ARM | PERCENT | 35%

Percent of samples with a deep deletion at this arm's locus. AUTHORED

values: 0.0 (722) Â· 0.0953 (75) Â· 0.1907 (32) Â· 0.286 (10) Â· 0.0954 (6) Â· 0.3813 (6) Â· 1.5253 (4) Â· 0.0955 (1)

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0

---

cnv_pct_gain
ARM | PERCENT | 35%

percent DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 16.11

> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€" âš multi-locus arms are DILUTED here*

---

cnv_pct_loh
ARM | PERCENT | 35%

percent DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 69.97

> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€" âš multi-locus arms are DILUTED here*

---

cnv_pct_loss
ARM | PERCENT | 35%

percent DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 11.34

> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€" âš multi-locus arms are DILUTED here*

---

cnv_sd_cn
ARM | REAL Â‰¥ 0 | 35%

standard deviation DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.9746

> *Defined on: arms with a mapped CN locus in the cohort stratum (856) â€" âš multi-locus arms are DILUTED here*

CNVC_ Â· 5

reference â€" context, provenance and rarely-quoted detail

> CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance.
>
> **5** columns Â· rung **arm** Â· coverage **31%â€"31%** (median 31%) Â· **4** share the block description
> 2× p-value Â· 2× signed âˆ'1â€¦1 Â· 1× count

cnvc_n
ARM | COUNT | 31%

count DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 1032

> *Defined on: arms with a CNâ†"expression concordance fit (756)*

---

cnvc_partial_p
ARM | P-VALUE | 31%

p-value DERIVED

example: hsa-let-7a-5p / let-7-5p/98-5p â†' 0.0001062

> *Defined on: arms with a CNâ†"expression concordance fit (756)*

cnvc_partial_q
ARM | P-VALUE | 31%

BH q for the arm's CN-coupling partial correlation. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.0006635`

---

cnvc_partial_rho
ARM | SIGNED Â˘'1Â€¦1 | 31%

Spearman Ï DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.1238`

> *Defined on: arms with a CNâ†"expression concordance fit (756)*

---

cnvc_spearman_rho
ARM | SIGNED Â˘'1Â€¦1 | 31%

Spearman Ï DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.0573`

> *Defined on: arms with a CNâ†"expression concordance fit (756)*

MODEL_ Â· 1

reference â€" context, provenance and rarely-quoted detail

1 columns Â· rung **arm** Â· coverage **30%â€"30%** (median 30%)
1× count

model_n_genes
ARM | COUNT | AGG EDGE | 30%

Which model variant produced this row. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 38`

> *Defined on: arms carrying >=1 curated HE edge in the model design (741)*

COMV_ Â· 2

reference â€" context, provenance and rarely-quoted detail

2 columns Â· rung **arm** Â· coverage **15%â€"15%** (median 15%)
1× count Â· 1× identifier

comv_module
ARM | COUNT | 15%

Which co-movement module this arm belongs to in the Â§6b-RETIRED heuristic lane. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 22`

---

comv_top_partner
ARM | IDENTIFIER | 15%

Co-expression module membership and top co-expressed partner. Pure co-expression â€" the cross-state class columns are deliberately excluded from this block. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' hsa-let-7f-5p(0.67);hsa-let-7b-5p(0.55);hsa-let-7e-5p(0.43);hsa-let-7b-3p(0.40);hsa-miR-101-3p.1(0.36)`

> *Defined on: arms in a co-expression module (371) â€" pure co-expression; the state-class columns are excluded*

CLOAD_ Â· 3

reference â€" context, provenance and rarely-quoted detail

Compartment LOADING of the arm.

3 columns Â· rung **arm** Â· coverage **8%â€"60%** (median 60%) Â· **2** share the block description
1× real â‰¥ 0 Â· 1× fraction 0â€"1 Â· 1× categorical

cload_dose_rpm
ARM | REAL Â‰¥ 0 | 60%

dose Â· reads per million DERIVED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 3.285e+04`

> *Defined on: arms with a compartment-loading row (1,473) â€" âš âš its source has NO PRODUCER (MH-111 ad-hoc)*

cload_lock

ARM | FRACTION 0Â€“1 | 8%

Is this arm's compartment loading LOCKED to one cell type rather than spread across the deconvolution? AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 0.256`

---

cload_top_compartment

ARM | CATEGORICAL | 60%

the top-ranked DERIVED

values: CAFs (1,335) Â· Endothelial (47) Â· PVL (35) Â· Myeloid (31) Â· Normal Epithelial (15) Â· B-cells (4) Â· Plasmablasts (3) Â· T-cells (3)

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' Endothelial`

> *Defined on: arms with a compartment-loading row (1,473) â€" âš âš its source has NO PRODUCER (MH-111 ad-hoc)*

HX_ Â· 1

reference â€" context, provenance and rarely-quoted detail

---

1 columns Â· rung **arm** Â· coverage **28%â€"28%** (median 28%)
1× real â‰¥ 0

hx_hly_baseline_imputed

ARM | REAL Â‰¥ 0 | 28%

â›" IMPUTED healthy baseline (miTED rank-transfer or seed-mate substitution) â€" LABELLED as imputed, never presented as a measurement. AUTHORED

example: `hsa-let-7a-5p / let-7-5p/98-5p â†' 12.32` seed_family_card.tsv

seed_family

One row per the seed family itself, gene-free. 83.7% of families are SINGLETONS â€” for those every 'family-level' statistic is trivially that one arm's value. Mask on `fam_single_member` first.

33 columns **1,959** rows **2** blocks median coverage **92%**

Contents â€” 2 blocks, most important first

SEED_ Â· 1

core â€” the columns findings rest on

1 columns Â· rung **key** Â· coverage **100%â€“100%** (median 100%)
1× identifier

seed_family
KEY IDENTIFIER 100%

KEY â€” the seed family itself, one row per family. AUTHORED

example: miR-4670-3p

FAM_ Â· 32

reference â€” context, provenance and rarely-quoted detail

Seed-family structure the arm sits in â€” membership, affinity and dominance within it.

32 columns Â· rung **family** Â· coverage **4%â€“100%** (median 91%) Â· **16** share the block description
12× count Â· 8× real â‰¥ 0 Â· 7× fraction 0â€“1 Â· 2× boolean Â· 1× categorical Â· 1× identifier Â· 1× signed âˆ’1â€¦1

fam_ctx_composition
FAMILY CATEGORICAL 29%

The family's genomic-context MIX as a tally string â€” `sense_coding_host:1` on 251 families, `sense_lncRNA_host:1` on 175, `antisense_overlap:1` next (29 distinct patterns). AUTHORED

example: miR-34b-5p/449c-5p â†’ sense_lncRNA_host:1,sense_coding_host:1

fam_ctx_context_homogeneous
FAMILY BOOLEAN 29%

Do ALL of this family's members share one genomic context â€” **True on 524 of 571 (91.8%)**? AUTHORED

values: True (524) Â· False (47)

example: miR-34b-5p/449c-5p â†’ False

fam_ctx_dose_coding_host_frac
FAMILY FRACTION 0â€“1 29%

dose Â· fraction DERIVED

example: miR-34b-5p/449c-5p â†’ 0

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

fam_ctx_dose_cotranscribed_frac
FAMILY FRACTION 0â€“1 29%

dose Â· fraction DERIVED

example: miR-34b-5p/449c-5p â†’ 0

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

fam_ctx_n_distinct_coding_hosts

FAMILY COUNT 29%

count DERIVED

values: 1.0 (288) Â· 0.0 (264) Â· 2.0 (17) Â· 3.0 (2)

example: miR-34b-5p/449c-5p â†’ 1

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_cur_degree_max

FAMILY COUNT 36%

maximum DERIVED

example: miR-3682-3p â†’ 1

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_dominant_arm

FAMILY IDENTIFIER 92%

the dominant one Â· per arm DERIVED

example: miR-4691-5p/6792-3p â†’ hsa-miR-4691-5p

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_dominant_share

FAMILY FRACTION 0â€“1 92%

The top member's share of the family's total dose. AUTHORED

example: miR-4691-5p/6792-3p â†’ 0.5101

---

fam_dose_hhi

FAMILY FRACTION 0â€“1 15%

Concentration of DOSE across the family's members. âœ… **FLOOR-CORRECTED â€” verified 2026-08-19**: its minimum sits far below the raw 1/k floor at every member count (0.0000 at k=2 where the raw floor is 0.500), and spearman(n_members, hhi) = **+0.1896** rather than the strongly negative value a raw index would give. AUTHORED

example: miR-378-3p â†’ 0.9346

---

fam_dose_max_log2

FAMILY REAL â‰¥ 0 92%

dose Â· maximum DERIVED

example: miR-4691-5p/6792-3p â†’ 0.4237

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_dose_med_log2

FAMILY REAL â‰¥ 0 92%

dose Â· median DERIVED

example: miR-4691-5p/6792-3p â†’ 0.4164

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_dose_total_log2

FAMILY REAL â‰¥ 0 100%

dose Â· total DERIVED

example: miR-4670-3p â†’ 0.3658

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_fame_npmid_sum

FAMILY COUNT 100%

sum DERIVED

example: miR-4670-3p â†’ 13

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_n_expressed

FAMILY COUNT 100%

count DERIVED

example: miR-4670-3p â†’ 1

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_n_guide

FAMILY COUNT 100%

count DERIVED

values: 0.0 (1,737) Â· 1.0 (188) Â· 2.0 (23) Â· 3.0 (3) Â· 6.0 (3) Â· 5.0 (2) Â· 4.0 (2) Â· 9.0 (1)

example: miR-4670-3p â†’ 0

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

fam_n_members

FAMILY COUNT 100%

count Â· family members DERIVED

example: miR-4670-3p â†’ 1

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_n_members_context

FAMILY COUNT 29%

count Â· family members DERIVED

values: 1.0 (486) Â· 2.0 (61) Â· 3.0 (14) Â· 5.0 (4) Â· 10.0 (2) Â· 4.0 (2) Â· 6.0 (1) Â· 8.0 (1)

example: miR-34b-5p/449c-5p â†’ 2

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_n_model_edges

FAMILY COUNT 100%

count DERIVED

example: miR-4670-3p â†’ 0

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_n_passenger

FAMILY COUNT 100%

count DERIVED

values: 0.0 (1,784) Â· 1.0 (167) Â· 2.0 (5) Â· 3.0 (3)

example: miR-4670-3p â†’ 0

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_n_shift_gt20_gated

FAMILY COUNT 100%

count Â· shift Â· gated DERIVED

values: 0.0 (1,911) Â· 1.0 (45) Â· 2.0 (3)

example: miR-4670-3p â†’ 0

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_real_n_c10_sum

FAMILY COUNT 100%

count Â· at the âˆ’0.10 threshold Â· sum DERIVED

example: miR-4670-3p â†’ 0

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_real_n_scored_sum

FAMILY COUNT 100%

count Â· sum DERIVED

example: miR-4670-3p â†’ 0

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_redund_m_eff

FAMILY REAL â‰¥ 0 4%

EFFECTIVE number of independent members after accounting for within-family correlation â€” the family's redundancy-corrected size (mode **2.0**, 50 distinct values). AUTHORED

example: miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 4.82

---

fam_redund_median_within_rho

FAMILY SIGNED âˆ’1â€¦1 4%

median Â· Spearman Ï DERIVED

example: miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 0.178

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_redund_overcount_frac

FAMILY FRACTION 0â€“1 4%

fraction DERIVED

example: miR-25-3p/32-5p/92-3p/363-3p/367-3p â†’ 0.035

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

---

fam_seed_shift_max_gated

FAMILY FRACTION 0â€“1 13%

shift Â· maximum Â· gated DERIVED

example: miR-27-3p â†’ 0.023

> *Defined on: seed families in the arm-card universe (1,959; â›” 83.7% single-member â‡’ shares/HHI are 1 or NaN)*

fam_single_member

FAMILY | BOOLEAN | 100%

ââ **THE DOMINANT FACT ABOUT THIS ENTIRE CARD: 1,639 of 1,959 families (83.7%) are SINGLETONS.** For those rows every 'family-level' statistic is TRIVIALLY that one arm's value, and `fam_dominant_share` is forced to exactly 1.0. AUTHORED

values: `True (1,639) Â· False (320)`

example: `miR-4670-3p â†' True`

---

fam_targetome_seed_med

FAMILY | REAL Â‰¥ 0 | 97%

median DERIVED

example: `miR-4688/6743-5p â†' 1888`

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_targetome_seq_med

FAMILY | REAL Â‰¥ 0 | 30%

median DERIVED

example: `miR-3615 â†' 2353`

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_var_hhi

FAMILY | FRACTION 0Â€"1 | 14%

Herfindahl concentration of `abund_sd` ACROSS the family's members â€" is the family's variance carried by one arm? AUTHORED

example: `miR-376b-5p/376c-5p â†' 6.327e-05`

---

fam_var_max

FAMILY | REAL Â‰¥ 0 | 89%

maximum DERIVED

example: `miR-4687-3p â†' 0.2317`

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*

---

fam_var_med

FAMILY | REAL Â‰¥ 0 | 89%

median DERIVED

example: `miR-4687-3p â†' 0.2317`

> *Defined on: seed families in the arm-card universe (1,959; â›" 83.7% single-member â‡' shares/HHI are 1 or NaN)*